

Major Exam

Engineering Mechanics MEL1010

29th November 2023

Total Marks: $5+5+4+6+5+5+5+5+16=56$

Total Questions: 9

Total Time: 2 hours (2:30 p.m. - 4:30 p.m.)

Question # 1, 2 and 5 have choices (OR)

Question # 9 (Theory questions)

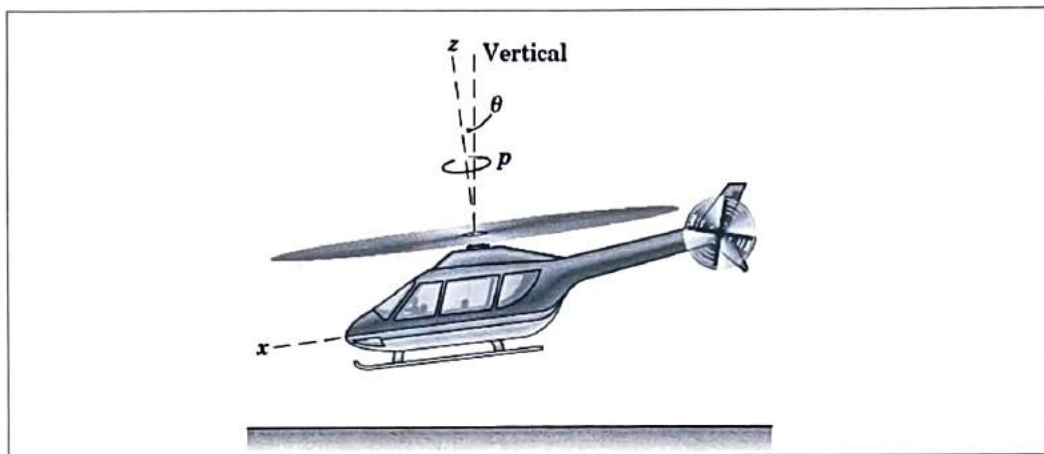
1 lb = 0.45 kg

1 mile = 1609.34 m

1 ft = 0.3048 m

Write your answer in SI units (if units are not mentioned)

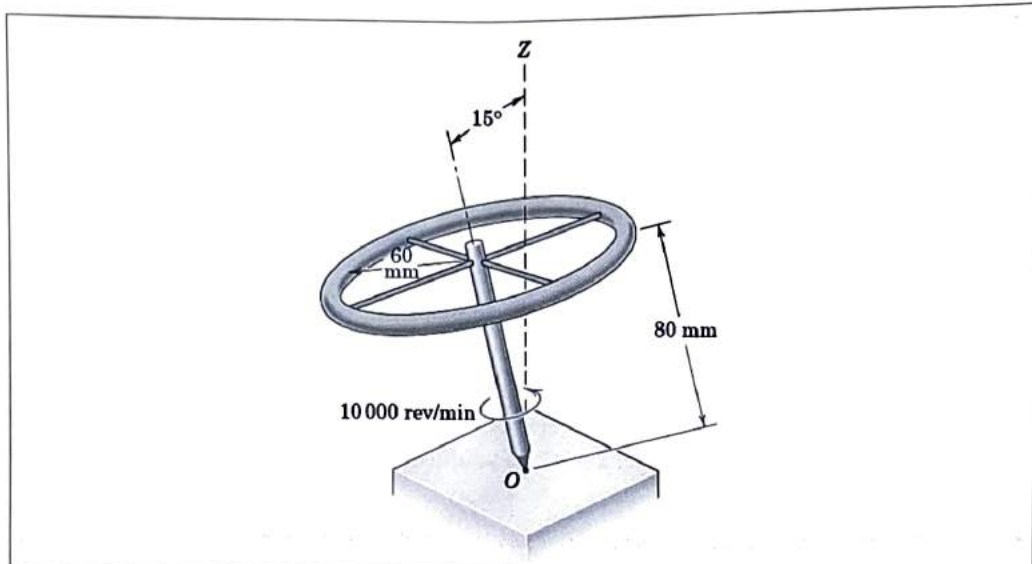
- 1) The blades and hub of the helicopter rotor weigh 140 lb (63.5 kg) and have a radius of gyration of 10 ft (3.048m) about the z-axis of rotation. With the rotor turning at 500 rev/min during a short interval following vertical liftoff, the helicopter tilts forward at the rate $\dot{\theta} = 10 \text{ deg/s}$ in order to acquire forward velocity.
- Determine the gyroscopic moment M transmitted to the body of the helicopter by its rotor.
 - Indicate whether the helicopter tends to deflect clockwise or counterclockwise, as viewed by a passenger facing forward. Explain your answer step by step.



OR

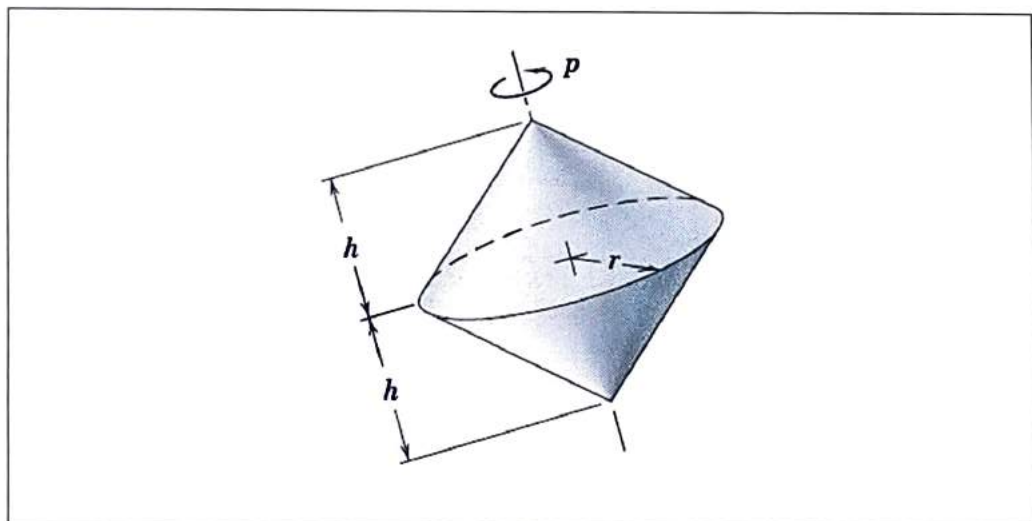
$$\alpha = \dot{m} \dot{\theta} \times \omega$$

- 1) A top consists of a ring of mass $m=0.52$ kg and mean radius $r=60$ mm mounted on its central pointed shaft with spokes of negligible mass. The top is given a spin velocity of 10 000 rev/min and released on the horizontal surface with the point O remaining in a fixed position. The axis of the top is seen to make an angle of 15° with the vertical as it precesses.
- Determine the number N of precession cycles per minute.
 - Identify the direction of the precession and sketch the body and space cones.



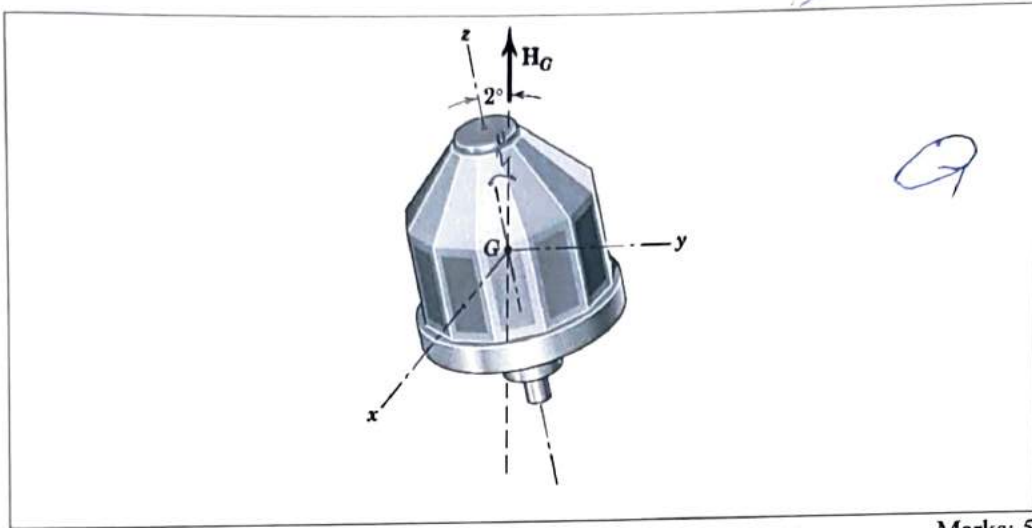
Marks: 5

- 2) The two solid cones with the same base and equal altitudes are spinning in space about their common axis at the rate p . For what ratio h/r will precession of their spin axis be impossible?



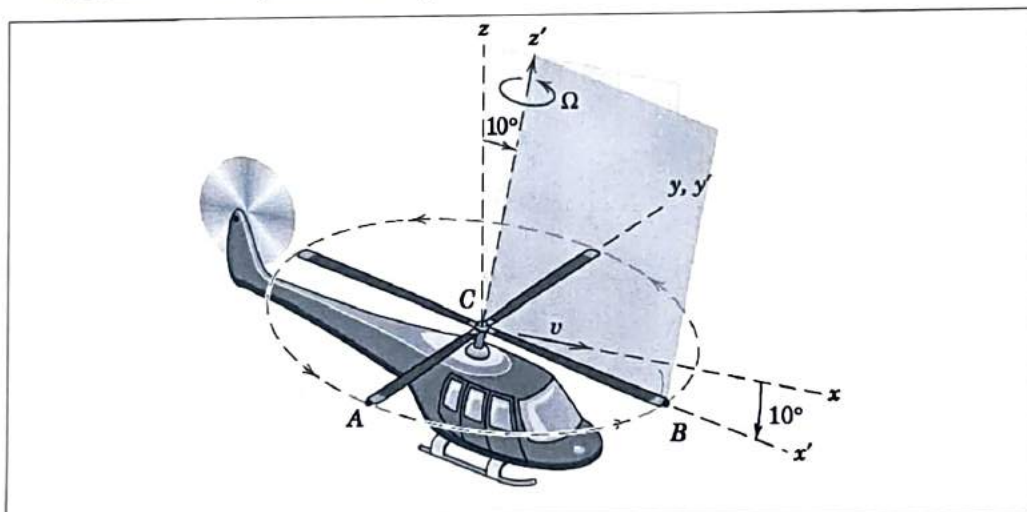
OR

- 2) The spacecraft shown is symmetrical about its z -axis and has a radius of gyration of 720 mm about this axis. The radii of gyration about the x - and y -axes through the mass center are both equal to 540 mm. When moving in space, the z -axis is observed to generate a cone with a total vertex angle of 4° as it precesses about the axis of total angular momentum. If the spacecraft has a spin velocity about its z -axis of 1.5 rad/s, compute the period τ of each full precession. Is the spin vector in the positive or negative z -direction?



Marks: 5

- 3) The helicopter is flying in the horizontal x -direction with a velocity $v = 120 \frac{\text{miles}}{\text{hr}} = 53 \text{ m/sec}$ and the plane of rotation of the 26-ft-diameter (7.9248m) rotor is tilted 10° from the horizontal x - y plane. The rotor blades rotate with an angular velocity $\Omega = 800 \text{ rev/min}$. For the instant represented write the vector expressions for the absolute velocities of rotor tip A and rotor tip B.

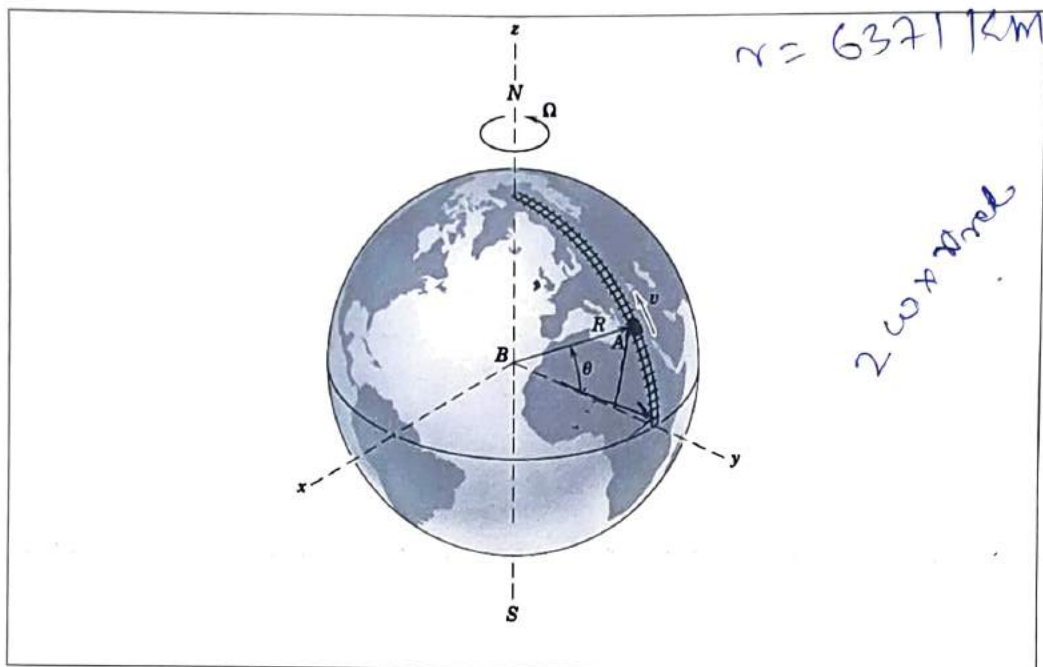


Marks: 4

4) An experimental vehicle A travels with constant speed v relative to the earth along a north-south track.

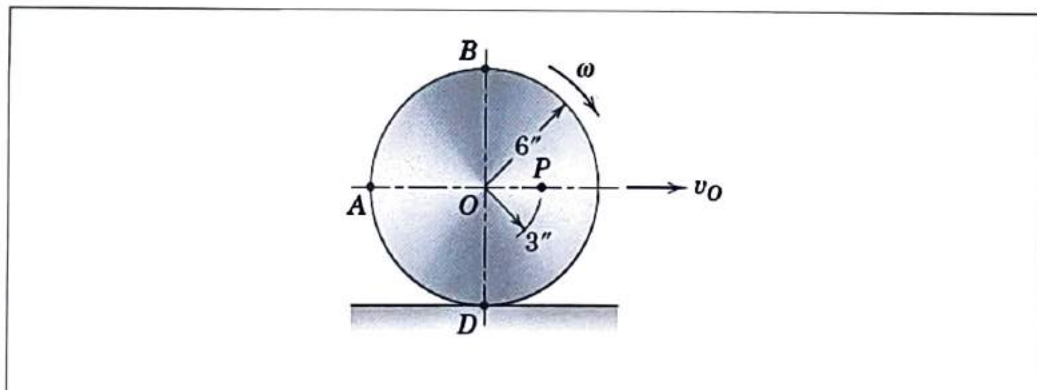
- Determine the Coriolis acceleration $\mathbf{a}_{cor}$ as a function of the latitude θ . Assume an earth-fixed rotating frame $Bxyz$ and a spherical earth.
- If the vehicle speed is $v = 500 \text{ km/hr}$, determine the magnitude of the Coriolis acceleration at (a) the equator and (b) the north pole.

Hint: One day = 24 hours = 24×60 minutes = $24 \times 60 \times 60$ seconds



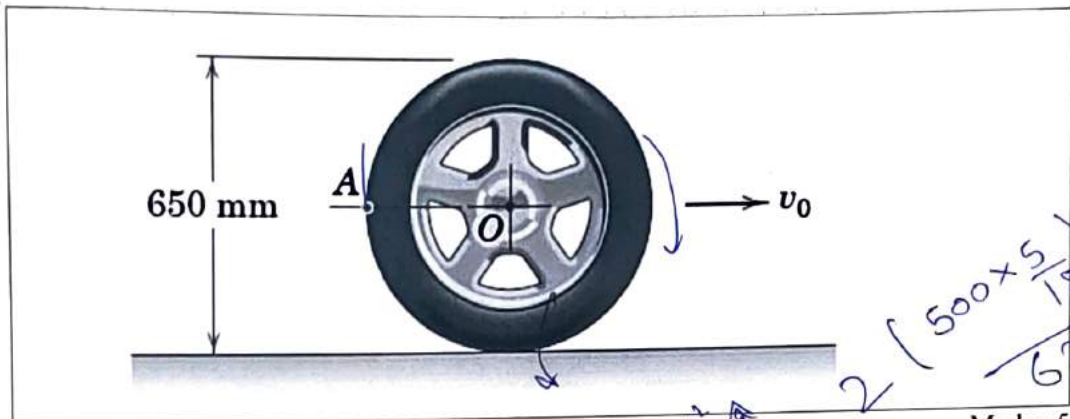
Marks: 6

5) The wheel slips as it rolls. If $v_o = 4 \text{ ft/sec}$ and if the velocity of A with respect to B is $3\sqrt{2} \text{ ft/sec}$, locate the instantaneous center C of zero velocity and find the velocity of point P.



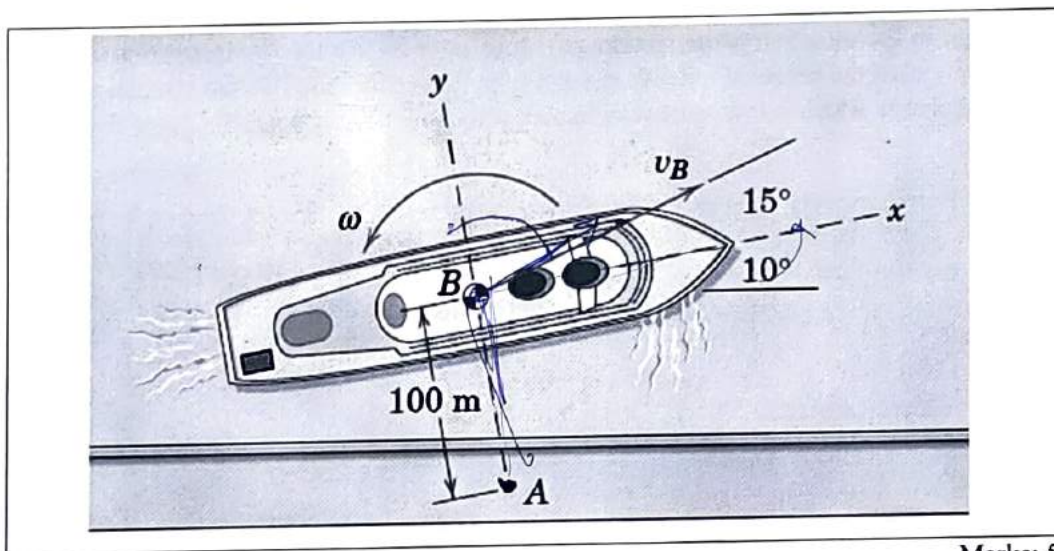
OR

- 5) The magnitude of the absolute velocity of point A on the automobile tire is 12 m/s when A is in the position shown. What are the corresponding velocity v_o of the car and the angular velocity ω of the wheel?
(The wheel rolls without slipping.)



Marks: 5

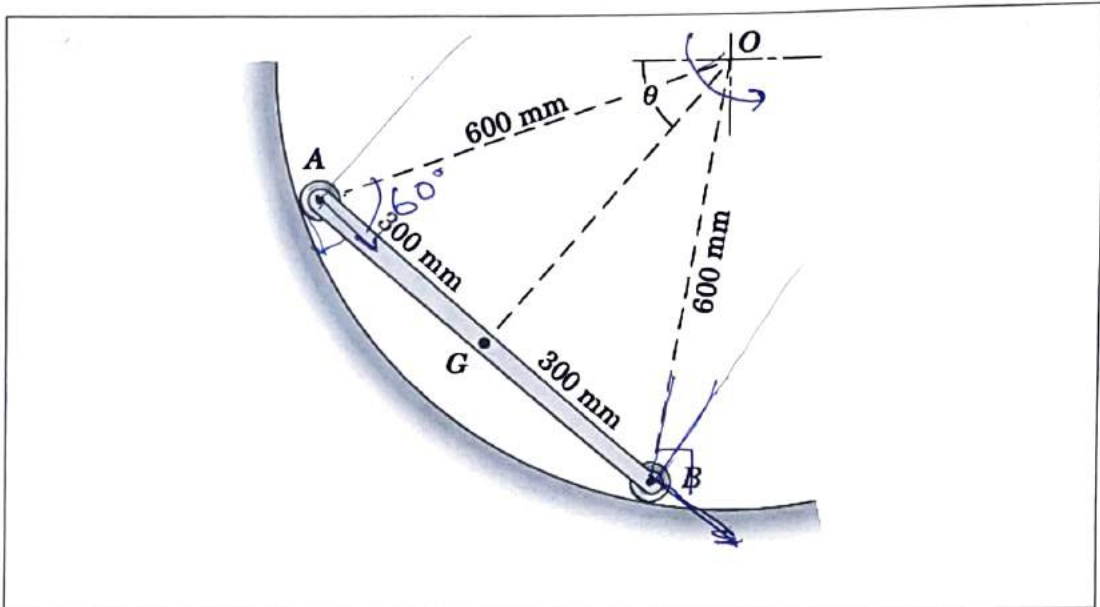
- 6) Under the action of its stern and starboard bow thrusters, the cruise ship has the velocity $v_B = 1 \text{ m/s}$ of its mass center B and angular velocity $\omega = 1 \text{ deg/s}$ about a vertical axis. The velocity of B is constant, but the angular rate ω is decreasing at 0.5 deg/s^2 . Person A is stationary on the dock. What velocity and acceleration of A are observed by a passenger fixed to and rotating with the ship? Treat the problem as two-dimensional.



Marks: 5

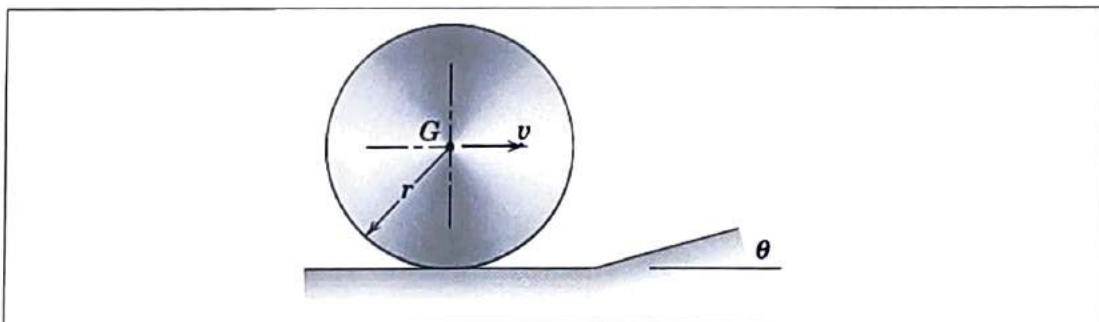
6.055 x 10^-2

- 7) The 24-kg uniform slender bar AB is mounted on end rollers of negligible mass and rotates about the fixed-point O as it follows the circular path in the vertical plane. The bar is released from a position which gives it an angular velocity $\omega = 2 \text{ rad/sec}$ as it passes the position $\theta = 45^\circ$. Calculate the forces F_A and F_B exerted by the guide on the rollers for this instant.



Marks: 5

- 8) A uniform circular disk which rolls with a velocity v without slipping encounters an abrupt change in the direction of its motion as it rolls onto the incline θ . Determine the new velocity v' of the center of the disk as it starts up the incline, and find the fraction n of the initial energy which is lost because of impact with the incline if $\theta = 10^\circ$.



$$\frac{10 \text{ R}}{36}$$

Marks: 5

9)

i. What are moments of inertia and products of inertia?

Marks: 1

ii. What are principal axes of inertia and principal moments of inertia?

Marks: 1

iii. What is inertial frame of reference and non-inertial frame of reference? Give examples

Marks: 1

iv. Explain body cone and space cone.

Marks: 1

v. Explain instantaneous center of zero velocity. Is it also instantaneous center of zero acceleration?

Marks: 1

vi. Explain instantaneous axis of rotation.

Marks: 1

vii. Explain body centrode and space centrode.

Marks: 1

viii. Explain Coriolis acceleration with an example.

Marks: 2

ix. Explain Work-Energy Principle.

Marks: 1

x. What is virtual work?

Marks: 1

xi. Explain how a gyroscope can be used for direction navigation systems.
[Hint: Explain the structure of Gyroscope (inner gimbal and outer gimbal). Explain **why** rotor axis always points in the same direction]

Marks: 2

xii. What is direct and retrograde precession? Explain with diagrams of body cone and space cone.

Marks: 3

Q1 Part 1 (Helicopter)

(5M)

Given.

$$m = 63.5 \text{ kg (140 lb)}$$

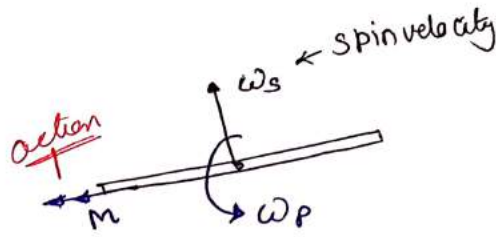
$$k = 3.048 \text{ m (10 ft)}$$

$$\omega_s = 500 \text{ rev/min}$$

$$= 500 \times \frac{2\pi}{60} = 52.4 \text{ rad/s}$$

$$\dot{\theta} = \omega_p = 10 \times \frac{\pi}{180} \text{ rad/s}$$

$$= 0.1745 \text{ rad/sec.}$$



ω_s : spin velocity

ω_p : precession velocity

$$M = I \bar{\omega}_p \times \bar{\omega}_s$$

$$= m k^2 \times \omega_p \omega_s$$

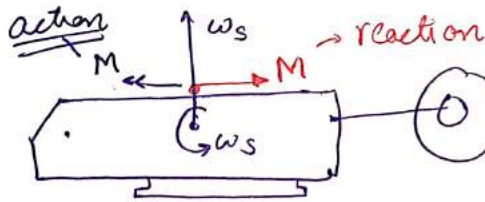
$$= 63.5 \times 3.048^2 \times 52.4 \times 0.1745$$

$$= 5394.2429 \text{ kg m}^2/\text{s}^2$$



Part B

CCW As



For FPS $m = \left(\frac{140 \text{ lbf}}{32.2} \right)$

$$m = \frac{140}{32.2} \times 10^2 \times 52.4 \times 0.1745 = 3970 \text{ lb-ft}$$

Force unit
FPS (lbf)
SI (N-m)

Q2 Part 1

5M

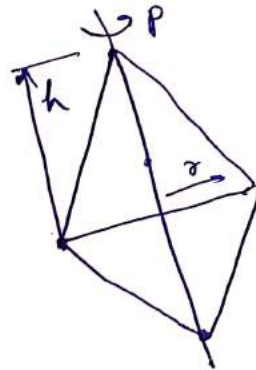
Q2

To find $(h/\hbar)$ at which precession of spin axis will be impossible,

~~Consider~~

Rate of precession.

$$\dot{\psi} = \frac{I_P}{(I_0 - I) \cos \theta}$$



for impossibility of ~~pre~~ precession

3 marks

$I_0 = I$
 $\uparrow$ about spin axis
 $\uparrow$ of precession

fill for this concept

Step 2 Calculation of I_0 & I .

For a cone:

$$I_{GG'} = \frac{3}{80} m (4r^2 + h^2)$$

$$I \rightarrow I_{xx} = \frac{3}{80} m (4r^2 + h^2) + \frac{mh^2}{16}$$

$$= \frac{3}{20} mr^2 + \frac{3mh^2}{80} + \frac{mh^2}{16}$$

$$= \frac{3}{20} mr^2 + \frac{8mh^2}{80} = \frac{mr^2}{20} + \frac{1}{10} mh^2 \quad (1)$$

$I_0 \rightarrow$

$$I_{zz} = \frac{3}{10} mr^2 \quad (2)$$

Equating (1) & (2)

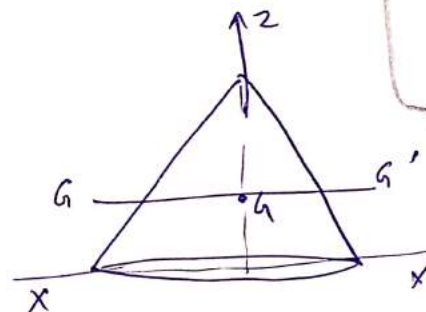
$$\frac{2 \times 3}{2 \times 10} mr^2 = \frac{3}{20} mr^2 + \frac{1}{10} mh^2$$

$$\frac{3}{20} mr^2 = \frac{1}{10} mh^2$$

$$\boxed{\frac{r}{h} = \sqrt{2/3}}$$

$$\boxed{\frac{h}{\hbar} = \sqrt{3/2}}$$

As



2 marks

Q5 Part 1

(5 M)

Rolling with slipping

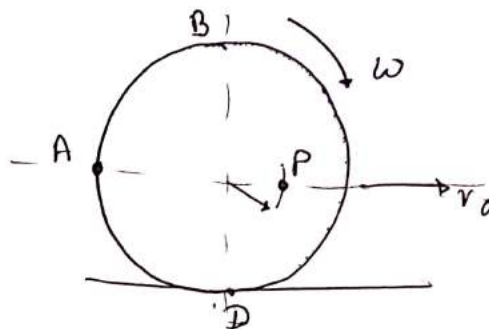
Q.5.1

Given, $|OP| = 3'' = 0.25 \text{ ft}$

$r_2 = 6'' = 0.5 \text{ ft}$

$V_0 = 4 \text{ ft/sec}$

$V_{A/B} = 3\sqrt{2} \text{ ft/sec}$



- To Find : (i) Locate the instantaneous Center C of zero velocity } -
 (ii) Find the velocity of point P

~~$V_{A/B} = 3\sqrt{2}$~~

$V_{A/B} = 3\sqrt{2} \text{ ft/sec}$

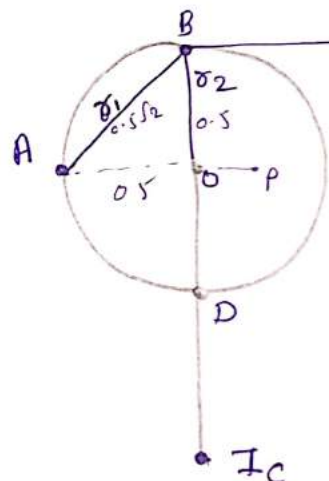
$$\Rightarrow 3\sqrt{2} = \omega r_2$$

$$\Rightarrow 3\sqrt{2} = \omega \times 0.5\sqrt{2}$$

$$\Rightarrow \omega = \underline{6 \text{ rad/sec}}$$

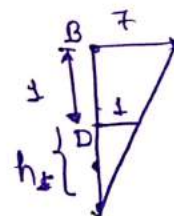
$$V_B = V_0 + \omega r_2$$

$$= 4 + 6 \times 0.5 = 4 + 3 = 7 \text{ ft/sec}$$



$$V_D = V_0 + \omega(-r_2)$$

$$= 4 - 6 \times 0.5 = 1 \text{ ft/sec}$$



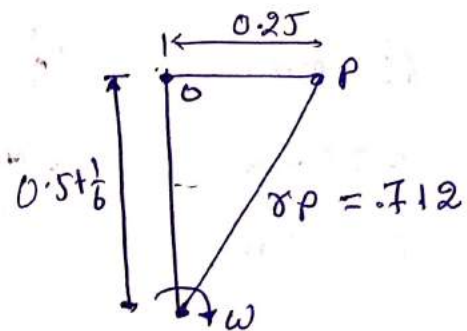
Finding/Calculation of I.C's

$$\frac{7}{1+h} = \frac{1}{h} \Rightarrow 6h = 1 \Rightarrow h = \frac{1}{6} = \frac{12}{6} = \underline{2''}$$

$$\therefore \boxed{h = 2''}$$

Calculation of V_P

$$\begin{aligned} V_P &= \cancel{V_I} + \omega r_P \\ &= 6 \times 0.712 \\ &= 4.27 \text{ m/sec } \underline{\underline{Ans}} \end{aligned}$$



Ans: (i) Location of I.C's

↳ IS at ~~20~~ 6" in the downward direction from O. (Center of circle)

(ii) $V_P = 4.27 \text{ m/sec } \underline{\underline{Ans}}$

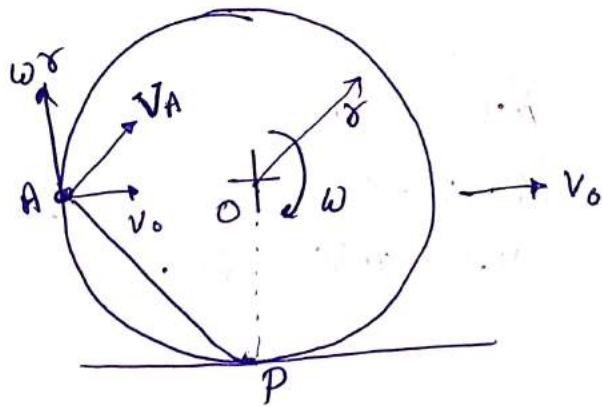
Q5 Part 2

(5M)

Q5

$$\begin{aligned} r &= \frac{650 \text{ mm}}{2} \\ &= 325 \text{ mm} \\ &= 0.325 \text{ m} \end{aligned}$$

$$v_A = 12 \text{ m/s}$$



$$v_A = \sqrt{v_0^2 + \omega^2 r^2}$$

$$\Rightarrow 12^2 = v_0^2 + \frac{v_0^2}{R^2} R^2$$

$$\omega = v_0 / R$$

$$\Rightarrow v_0 = \frac{12}{\sqrt{2}} = 8.48528 \text{ m/sec}$$

$$\omega = \frac{v_0}{R} = \frac{8.48528}{0.325} = \underline{26.11 \text{ rad/sec}}$$

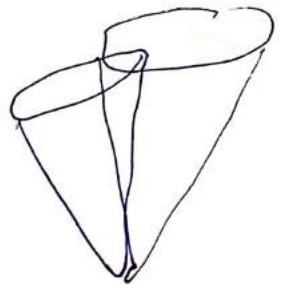
$$\dot{\psi} = \frac{I p}{(I_0 - I) \cos \theta} \approx 7/30$$

$p = 1.5 \text{ rad/s}$ → ~~spin~~ precession

precession
spin
~~precession~~

$$\dot{\psi} = \frac{m k^2 p}{(m k_0^2 - m k^2) \cos 2^\circ}$$

$$\dot{\psi} = \frac{p}{\left(\frac{k_0^2}{k^2} - 1\right) \cos 2^\circ} \Rightarrow \frac{1.5}{\left(\frac{0.72^2}{0.54^2} - 1\right) \cos 2^\circ}$$



$$\frac{1.5}{\left(\frac{0.54^2}{0.72^2} - 1\right) \cos 2^\circ}$$

Part 2 Q27

$$\dot{\psi} = \frac{I p}{(I_0 - I) \cos \theta} \quad \text{--- ②}$$

$\dot{\psi}$ = if all substitution correct with correct calculation.

where,

p = spin

I_0 = MoI of precession

I = about spin (z-axis)

$$\dot{\psi} = \frac{m k^2 p}{(m k_0^2 - m k^2) \cos \theta}$$

$$\dot{\psi} = \frac{p}{\left(\left(\frac{k_0}{k}\right)^2 - 1\right) \cos \theta}$$

$$\theta = 2^\circ$$

$$\dot{\psi} = 3.44 \text{ rad/s}$$

$$t = \frac{2\pi}{\dot{\psi}} = 1.83 \text{ sec}$$

③

(-ve z dir)



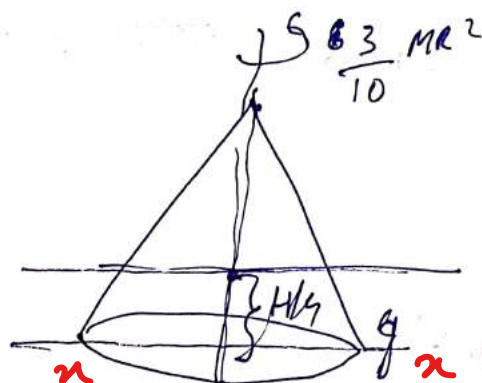
Book Page.
Ref. 565

Q2

$$MOI = \frac{3}{10} MR^2$$

$$= \frac{3}{10} MR^2$$

Calculation
of I_{xx}



$$I_{xx} = \frac{3}{80} m(4r^2 + h^2) + m \frac{h^2}{16}$$

$$\frac{3mh^2}{80} + \frac{mh^2}{16}$$

$$\frac{3+5}{80} = \frac{8mh^2}{80}$$

$$I_{xx} = \frac{3}{80} m(4r^2 + h^2) + m \frac{h^2}{16}$$

$$= \frac{3mr^2}{20} + \frac{3mh^2}{80} + \frac{mh^2}{16}$$

$$= \frac{3mr^2}{20} + \frac{3mh^2 + 5mh^2}{80}$$

$$= \frac{3mr^2}{20} + \frac{1}{10} mh^2$$

Q6

①

Q6

'B' is the center of mass of the ship

Given, Velocity of ship, $V_B = 1 \text{ m/s}$

5 marks

Angular velocity, $\omega = 1 \text{ deg/sec} = \frac{\pi}{180} \text{ rad/sec}$.

Angular acceleration, $\alpha = -0.5 \text{ deg/s}^2 = -0.5 \frac{\pi}{180} \text{ rad/s}^2$.

Person A is stationary on the dock.

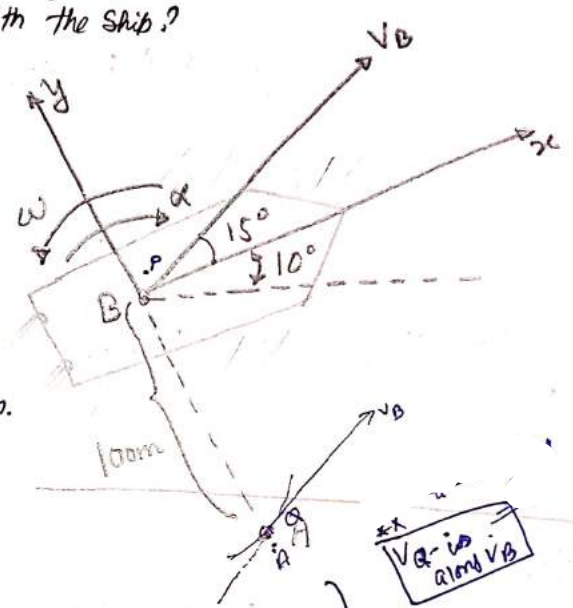
To find:- Velocity and acceleration of A relative to ~~the~~ a passenger fixed to and rotating with the ship?

i.e. $V_{rel} = ?$ $a_{rel} = ?$

Solve:

Assumption

Passenger/Person A is at Point B, or its distance from B is very small relative to distance, $BA = 100 \text{ m}$.



M-I

$$\vec{V}_A = \vec{V}_B + \vec{V}_{A/B} \quad \text{--- (1)}$$

$$\Rightarrow \vec{V}_A = \vec{V}_B + (\vec{\omega} \times \vec{r} + \vec{V}_{rel})$$

$$\Rightarrow 0 = \vec{V}_B + \vec{\omega} \times \vec{r} + \vec{V}_{rel}$$

$$\Rightarrow \vec{V}_{rel} = -(\vec{V}_B + \vec{\omega} \times \vec{r}) \quad \text{--- (2)}$$

$$\begin{aligned} \vec{V}_B &= \cos 15^\circ \hat{i} + \sin 15^\circ \hat{j} \\ &= -1.7453 \cdot 0.966 \hat{i} + 0.26 \hat{j} \quad \text{--- (3)} \end{aligned}$$

$$\vec{\omega} \times \vec{r} = \frac{\pi}{180} \times 100 (\hat{k} \times \hat{j}) = 1.7453 \hat{i} \quad \text{--- (4)}$$

Substituting (3) & (4) into (2) yields.

$$\vec{V}_{rel} = (-1.7453 - 0.966) \hat{i} - 0.26 \hat{j} = -2.7113 \hat{i} - 0.26 \hat{j} \text{ m/s}$$

$$\vec{V}_{A/B} = \vec{V}_A / \alpha + \vec{V}_{A/B}$$

$$\vec{V}_{A/B} = \vec{V}_{rel}$$

$$\vec{V}_{A/B} = \vec{\omega} \times \vec{r}$$

Q. is a point fixed with x-y & coinciding with A.

Calculation of relative acceleration (a_{rel})

$$\vec{a}_A = \vec{a}_B + \vec{a}_{A/B}$$

$$= \vec{a}_B + \vec{a}_r + \vec{a}_t + \vec{a}_{cor} + a_{rel}$$

$$= \vec{a}_B + [\vec{\omega} \times (\vec{\omega} \times \vec{r}) + (\vec{\alpha} \times \vec{r}) + 2\vec{\omega} \times \vec{v}_A] + \vec{a}_{rel}$$

$$[\vec{a}_A = 0, \vec{a}_{rel} = ? , \vec{a}_B = 0]$$

$$\vec{a}_{rel} = -[\vec{\omega} \times (\vec{\omega} \times \vec{r}) + \vec{\alpha} \times \vec{r} + 2\vec{\omega} \times \vec{v}_B]$$

Form Eq. (3)

$$= -\left[(\omega^2 r) \hat{j} + \omega(\alpha r) \hat{i} + 2 \times \frac{\pi}{180} (kr) \times (0.966 \hat{i} + 0.263 \hat{j}) \right]$$

$$= \left[\left(\frac{\pi}{180} \right)^2 \times 100 \hat{j} + 0.5 \frac{\pi}{180} \times 100 \hat{i} + 0.3372 \hat{j} - 9.033 \times 10^3 \hat{i} \right]$$

$$a_{rel} = \left[(0.0641817) \hat{j} + 0.8636 \hat{i} \right] \underline{\underline{m/s^2}}$$

Ans.

M-II

Consider the phenomenon relative to a passenger sitting inside the ship.

Person B will see that A is ~~revolving~~ rotating at ω in clockwise sense and α is in clockwise.

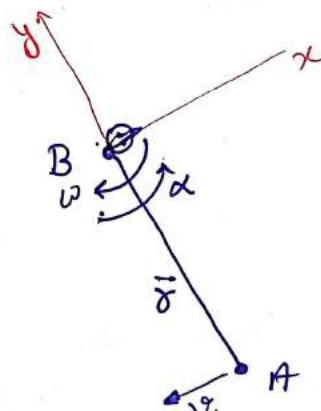
Thus:

$$\vec{v}_{A/B} =$$

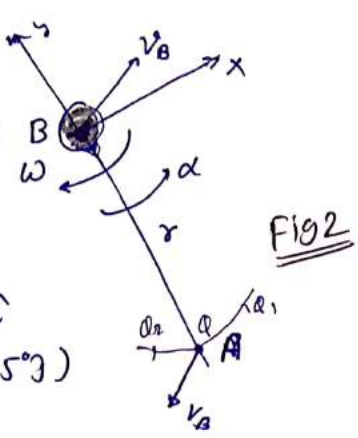
And it will also

observe that point A

is moving in (-x) dirⁿ @ speed v_B



Thus, $\vec{V}_{rel} = \vec{\omega} \times \vec{r}$ Let point is fixed To frame $x_1 y_1 z_1$ or coinciding with A at this moment

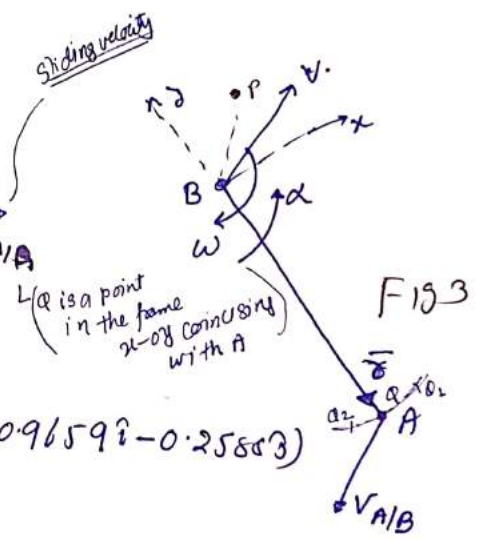


$$\begin{aligned} \vec{V}_{rel} &= \vec{\omega} \times \vec{r} + \vec{V}_{A/A} \\ &= \left(-\frac{\pi}{180} \times 100\right) \hat{i} + (-\cos 15^\circ \hat{i} - \sin 15^\circ \hat{j}) \\ &= -1.7453 \hat{i} - 0.966 \hat{i} - 0.26 \hat{j} \end{aligned}$$

$$\therefore \vec{V}_{rel} = \underline{-2.7113 \hat{i} - 0.26 \hat{j} \text{ m/sec}}$$

Calculation of Accn

$$\vec{a}_{rel} = \omega \times (\omega \times \vec{r}) + (\alpha \times \vec{r}) + 2 \vec{\omega} \times \vec{V}_{A/A}$$



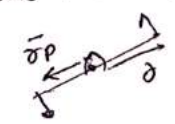
$$\begin{aligned} &= \omega^2 r \hat{j} + \alpha r \hat{i} \\ &\quad + \frac{2\pi}{180} (-\hat{k}) (-0.9659 \hat{i} - 0.2588 \hat{j}) \\ &= \left(\frac{\pi}{180}\right)^2 \times 100 \hat{j} + 0.5 \frac{\pi}{180} \times 100 \hat{i} + 0.3372 \hat{j} - 9.033 \times 10^{-3} \hat{i} \end{aligned}$$

$$\therefore a_x = 0.5 \frac{\pi}{180} \times 100 - 9.033 \times 10^{-3} \hat{i} = \underline{0.8636 \text{ m/s}^2}$$

$$a_y = 0.03372 + \left(\frac{\pi}{180}\right)^2 \times 100 = \underline{0.0641817 \text{ m/s}^2}$$

If: passenger is at location $\vec{r}_p$ from B

→ Then in the calculation of velocity Take this into account as $\vec{\omega} \times \vec{r}_p$ added with $\vec{V}_{rel}$



Q for acc?

Two extra component comes:

- ① $a_{tan} = \alpha \times \vec{r}_p$
- ② $a_c = \vec{\omega} \times (\vec{\omega} \times \vec{r}_p)$

Q7

5M

17

Given, $\omega = 2 \text{ rad/s}^2$

$$\begin{aligned} d_2 &= b \times \cos 30^\circ \\ &= 0.6 \times \frac{\sqrt{3}}{2} \\ &= 0.3\sqrt{3} \text{ m} \end{aligned}$$

$$d_1 = 0.6 \text{ m.}$$

To find Normal reaction F_A & F_B

Step 1: using the Newton's 2nd Law for dynamic equation.

Along y-dirn

$$\sum F_y = m a_y \quad - (1)$$

$$\Rightarrow (F_A \cos 30^\circ + F_B \cos 30^\circ) - (mg \cos 45^\circ) = m \omega^2 d_2$$

$$\Rightarrow \frac{F_A \sqrt{3}}{2} + \frac{F_B \sqrt{3}}{2} = m \left(\omega^2 d_2 + \frac{g}{\sqrt{2}} \right) \quad - (2)$$

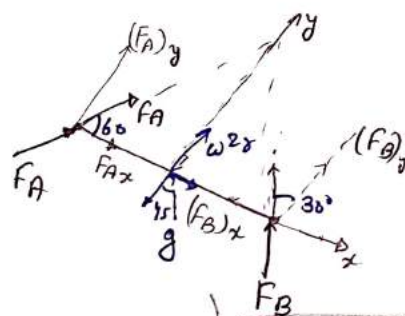
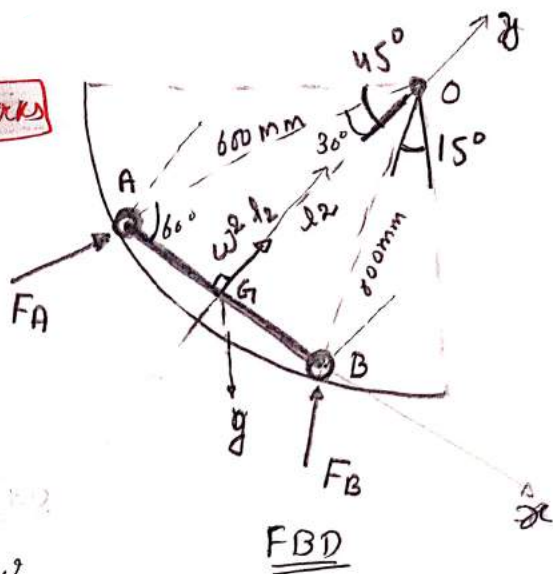
Along x-dirn

$$\sum F_x = m a_x$$

$$\Rightarrow \frac{F_A}{2} + \left(-\frac{F_B}{2} \right) + mg \cos 45^\circ = m a_{cm} \quad - (3)$$

Now Step 2 Calculation of a_{cm}

5 marks



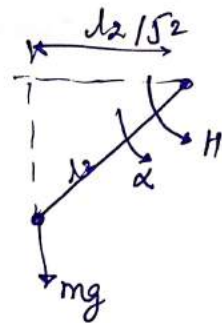
Calc. of a_{com}

Torque applied by gravity force.

$$a_{com} = \alpha r$$

$$= \frac{\sum H}{I} r$$

$$= \left(\frac{mg \times l_2 / \sqrt{2}}{m l_2^2 + \frac{m L^2}{12}} \right) l_2$$



$$L = 0.6 \text{ m} \\ = \text{length of rod.}$$

$$= \frac{9.81 \times (0.3\sqrt{3})^2 / \sqrt{2}}{(0.3\sqrt{3})^2 + 0.6^2 / 12}$$

$$a_{com} = 6.243045 \text{ m/s}^2. \text{ Ans} - (4)$$

Substituting (4) into (3), yields:

$$\frac{F_A}{2} - \frac{F_B}{2} + \frac{mg}{\sqrt{2}} = m \times 6.243045 - (5)$$

add (from 5)

$$\frac{F_A}{2} + \frac{F_B}{2} = m (\omega^2 l_2 + g / \sqrt{2}) \frac{1}{\sqrt{3}} - (6)$$

+ adds (5), (6) yields,

$$F_A = m \left[\frac{\omega^2 l_2}{\sqrt{3}} + \frac{g}{\sqrt{6}} + 6.243045 - g / \sqrt{2} \right]$$

$$= m \times \left[4 \times \frac{0.3\sqrt{3}}{\sqrt{3}} + \frac{9.81}{\sqrt{6}} + 6.243045 - \frac{9.81}{\sqrt{2}} \right]$$

$$= m \times 4.5112 = 108.269 \text{ N. (m = 24 kg)} \text{ Ans}$$

and

$$F_B = F_A + \frac{mg}{\sqrt{2}} - 2m \times 6.243045$$

$$= 108.269 + 48 \left[\frac{9.81}{\sqrt{2}} - 6.243045 \right] = 141.566 \text{ N Ans}$$

Q 8

Q8 Given.

Rolling without Slipping

5 marks

Initial velocity, $= V_1$

Final velocity $= V_2$

Ramp inclination $= \theta$

Radius $= r$

To find : i) V_2 just after the impact

ii) Fraction of initial energy lost due to impact.

$h = r \cos \theta$ — (1)

Fig. 3 or 2

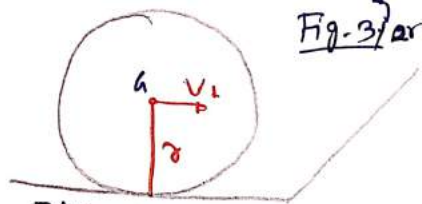


Fig1 Before Impact

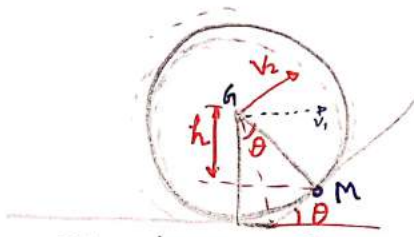


Fig2. just after impact

Principle used:

Conservation of Angular momentum @ point of impact.

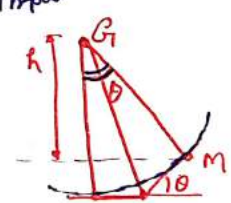


Fig. 3 are of Circle

Let M be the Point of impact,

$h = r \cos \theta$

Case I:

Initial Angular momentum about M

$H_i = H_G + \vec{r} \times \vec{G}_i$ — (2)

angular momentum about C.O.G. Linear momentum
arm length from the point of impact

$$H_i = \left(\frac{mr^2}{2} \right) \frac{V_1}{r} + \vec{r}_i \times m\vec{V}_1$$

$$= \frac{mr^2}{2} \frac{V_1}{r} + m V_1 r \cos \theta$$
 — (3)

$$H_G = I_G \omega_i$$

$$= \frac{mr^2}{2} \cdot \frac{V_1}{r}$$

Q 8 Contd.

Case II Just after the impact

$$\begin{aligned} H_f &= I_G \omega_2 + \bar{r} \times G_2 \\ &= \frac{m \bar{r}^2}{2} \frac{V_2}{\bar{r}} + m \bar{r} V_2 \quad - (3) \end{aligned}$$

Equating ① & ② yield

$$\frac{\bar{r}^2 V_1}{2 \bar{r}} + V_1 \bar{r} \cos \theta = \frac{\bar{r}^2}{2} \frac{V_2}{\bar{r}} + \bar{r} V_2$$

$$\Rightarrow \frac{V_1 \bar{r}}{2} + V_1 \bar{r} \cos \theta = \frac{\bar{r} V_2}{2} + \bar{r} V_2 = \frac{3}{2} V_2 \bar{r}$$

$$\Rightarrow \boxed{V_2 = \frac{2}{3} V_1 (0.5 + \cos \theta)} \quad - (5)$$

Ans

Part 2 Calculation of Energy lost

$$\frac{\Delta E}{E_i} = \frac{\frac{1}{2} I (\omega_i^2 - \omega_f^2)}{\frac{1}{2} I \omega_i^2} = 1 - \frac{\omega_f^2}{\omega_i^2}$$

$$\omega = \frac{V}{r}$$

$$= 1 - \frac{V_2^2}{V_1^2} = 1 - \left\{ \frac{2}{3} (0.5 + \cos \theta) \right\}^2$$

$$= 1 - \frac{4}{9} (0.5 + \cos 10^\circ)^2$$

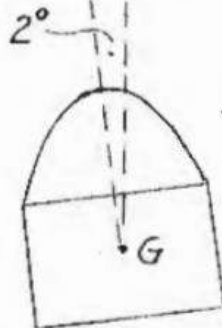
$$= 0.020153 \quad \underline{\underline{Ans}}$$

Part 02

Q.25

For zero moment Eq. 7/30 is

$$\dot{\psi} = \frac{I p}{(I_0 - I) \cos \theta} = \frac{p}{\left(\frac{k_0^2}{\bar{k}^2} - 1\right) \cos \theta}$$



where $\bar{k} = 0.72 \text{ m} \rightarrow$ about z

$k_0 = 0.54 \text{ m} \rightarrow$ about x, y .

$p = 1.5 \text{ rad/s} \rightarrow$ Spin velocity.

$\theta = 2^\circ$

$(I = \bar{k}^2 m) > (I_0 = k_0^2 m)$ so retrograde precession with p in negative z -dir. }

$$\begin{aligned} \text{Period } T &= \left| \frac{2\pi}{\dot{\psi}} \right| = 2\pi \left| \frac{(k_0^2/\bar{k}^2 - 1)}{p} \cos \theta \right| \\ &= 2\pi \left| \frac{(0.54/0.72)^2 - 1}{1.5} \cos 2^\circ \right| = \underline{1.831 \text{ s}} \end{aligned}$$

Q1.2

$$I_{zz} = mr^2, \quad k = r = 0.060 \text{ m}$$

$$p = 10\,000 (2\pi/60) = 1047 \text{ rad/s}$$

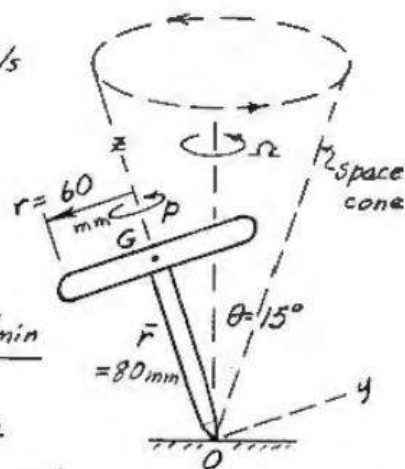
From Eq. 7/25,

$$\Omega \approx \frac{g \bar{r}}{k^2 p} = \frac{9.81(0.080)}{(0.060)^2 (1047)}$$

$$= 0.208 \text{ rad/s}$$

$$N = \frac{\Omega}{2\pi} 60 = \frac{0.208}{2\pi} \times 60 = \underline{1.988 \text{ cycles/min}}$$

With $\Omega = \dot{\psi}$ very small, the body cone is too small to observe, so space cone is the only relatively apparent cone.



(Note direction of precession on diagram.)

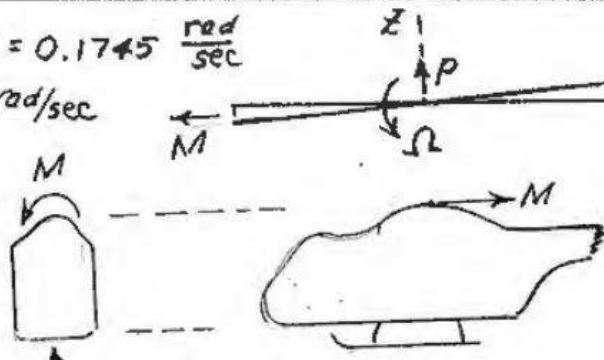
Q 1.1

$\Omega = \frac{10}{180} \pi = 0.1745 \frac{\text{rad}}{\text{sec}}$

$p = \frac{500}{60} 2\pi = 52.4 \frac{\text{rad}}{\text{sec}}$

$I = \frac{140}{32.2} 10^2 = 435 \frac{\text{lb-ft-sec}^2}{\text{lb-ft-sec}^2}$

$M = I \Omega p$
 $= 435 (0.1745) 52.4$
 $= 3970 \text{ lb-ft}$



As viewed by passenger looking forward

Conclusion: CCW deflection

4 Marks

③

Q3

$$\text{Velocity, } v = 120 \text{ miles/hr} = 53 \text{ m/s}$$

$$\text{Angular velocity, } \omega = 800 \text{ rev/min} = 800 \times \frac{2\pi}{60} \text{ rad/s} = 83.776 \text{ rad/s}$$

$$\omega \text{ in term of vector exp.} = 83.776 [\sin 10^\circ \hat{i} + \cos 10^\circ \hat{k}] \rightarrow$$

$$r_{CA} = -3.9624 \hat{j}, \quad r_{CB} = [\cos 10^\circ \hat{i} - \sin 10^\circ \hat{k}] \cdot 3.9624$$

Velocity at A w.r.t C in terms of v_c

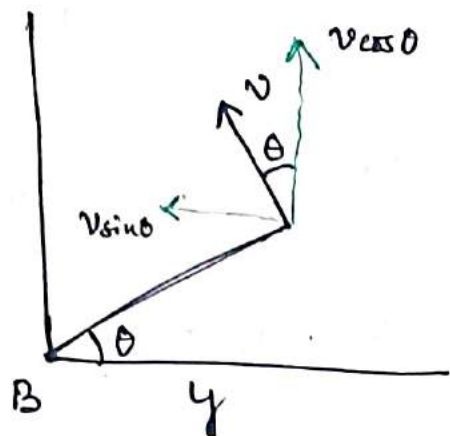
$$\begin{aligned} (*) \quad v_A &= v_c + \omega \times r_{CA} \\ &= 53 \hat{i} + 83.77 [\sin(10^\circ) \hat{i} + \cos(10^\circ) \hat{k}] \times (-3.9624 \hat{j}) \\ &= 53 \hat{i} + 83.77 [0.173 \hat{i} + 0.984 \hat{k}] \times (-3.9624 \hat{j}) \\ &= 53 \hat{i} + [14.492 \hat{i} + 82.429 \hat{k}] \times (-3.9624 \hat{j}) \\ &= 53 \hat{i} - 57.423 \hat{k} + 326.61 \hat{i} \end{aligned}$$

$$v_A = 379.616 \hat{i} - 57.423 \hat{k} \text{ m/s.} \rightarrow$$

$$\begin{aligned} (*) \quad v_B &= v_c + \omega \times r_{CB} \\ &= 53 \hat{i} + 83.77 [\sin(10^\circ) \hat{i} + \cos(10^\circ) \hat{k}] \times 3.9624 [\cos(10^\circ) \hat{i} - \sin(10^\circ) \hat{k}] \\ &= 53 \hat{i} + (83.77)(3.9624) [\sin^2(10^\circ) + \cos^2(10^\circ)] \hat{j} \end{aligned}$$

$$v_B = 53 \hat{i} + 331.954 \hat{j} \rightarrow$$

(4)

Q46 Marks

(6)

Coriolis acceleration, $a_c = 2v\omega = 2v\omega \sin \theta$

$$= 2\omega \times v_{rel}$$

$$v_{rel} = v [-\sin \theta j + \cos \theta k]$$

$$\therefore \textcircled{i} a_c = 2\omega k \times v [-\sin \theta j + \cos \theta k]$$

$$a_c = 2\omega v \sin \theta i$$

ii) At equator, $\theta = 0^\circ$

$$\textcircled{*} \underline{a_c} = 2\omega v \times 0 = 0$$

b) At N-pole, $\theta = 90^\circ$

$$\omega_{Earth} = \frac{2\pi}{24 \times 60 \times 60} = 7.27 \times 10^{-5} \text{ rad/s}$$

$$v = 500 \times \frac{5}{18} = 139 \text{ m/s}$$

$$\textcircled{*} \underline{a_{c, pole}} = 2 \times 7.27 \times 10^{-5} \times 139 = 0.0202 \text{ m/s}^2$$

(1)

Duration: 120 mins

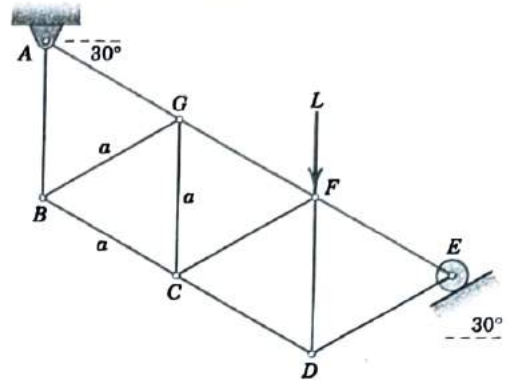
Maximum marks: 60

Instruction:

1. There are total 4 pages in the question paper make sure they are printed.
2. Use of Scientific calculators are allowed.
3. Take suitable assumption whenever necessary.
4. Consider the nature of the question asked (vector, scalar and units) and answer accordingly.
5. Marks will be given for Final Answer not the choice of method and steps. (Step marking, if any, will be given till the first mistake was made.)
6. You are allowed to use any method or approach unless specified in question.
7. Question paper is printed in bilingual format (English and Hindi); however, in case of any discrepancy, matter written in English is considered as final.

1. The truss is composed of equilateral triangles of side a , is supported, and loaded as shown. Determine the reaction at point E and the forces in member BC and CG. (1+3+3 = 7 marks)

ट्रस भुजा a के समबाहु त्रिभुजों से बना है, समर्थित है और दिखाए गए अनुसार लोड किया गया है। बिंदु E पर प्रतिक्रिया और भाग BC और CG में बल निर्धारित करें। (1+3+3 = 7 marks)



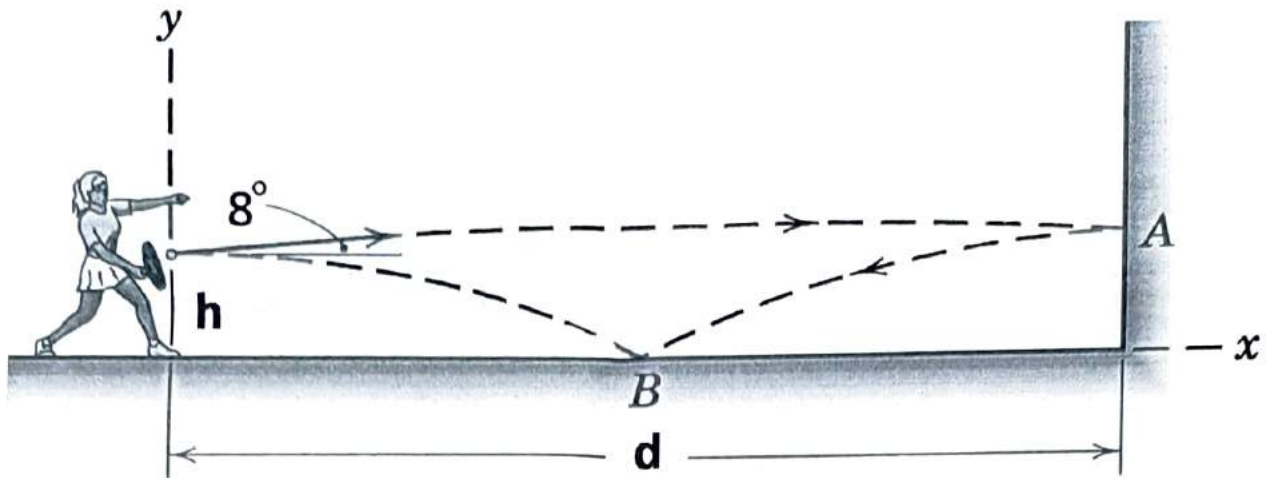
2. The squash player practices by hitting the ball against the wall at A. The ball bounces off the court surface at B and then reaches to same point from where player hit it every time. If the player has hit the ball with velocity $v_0 = 25 \text{ m/s}$ and at an angle $\theta = 8^\circ$ with respect to horizontal, from the height of 1 meter from floor and 10 meters away from the wall. The mass of ball is 25 gm, and the time of impact is 0.02 sec. Considering that when ball reaches to the player, it only has velocity in horizontal direction: (8 marks)

- Find the coefficient of restitution for wall and the floor which are related as the ratio of coefficient of restitution of wall to coefficient of restitution of floor is equal to coefficient of restitution of floor itself. (1 marks)
- Compute the horizontal and vertical component of velocity before and after the impact for both impact at A and impact at B. (2.5 marks)
- The height at which the ball hits the wall. (0.5 Marks)
- Recovery time (time between two consecutive hits by player during practice) (2 marks)
 - The horizontal and vertical component of reaction R (Between ball and racket), magnitude of reaction force R and angle made by reaction R with horizontal. (2 marks)

स्कॉश खिलाड़ी A पर दीवार के खिलाफ गेंद को मारकर अभ्यास करता है। गेंद B पर कोर्ट की सतह से उछलती है और फिर उसी बिंदु पर पहुंचती है जहां से खिलाड़ी हर बार इसे मारता है। यदि खिलाड़ी ने गेंद को $v_0 = 25 \text{ m/s}$ वेग से और क्षैतिज के संबंध में $\theta = 8^\circ$ के कोण पर, फर्श से 1 मीटर की ऊंचाई से और दीवार से 10 मीटर की दूरी पर मारा है। गेंद का द्रव्यमान 25 ग्राम है, और प्रभाव का समय 0.02 सेकंड है। यह मानते हुए कि जब गेंद खिलाड़ी के पास पहुंचती है, तो गेंद का वेग केवल क्षैतिज दिशा में होता है: (8 marks)

- दीवार और फर्श के लिए पुनर्स्थापन का गुणांक ज्ञात करें, यदि दीवार के पुनर्स्थापन के गुणांक और फर्श के पुनर्स्थापन के गुणांक का अनुपात, फर्श के पुनर्स्थापन के गुणांक के बराबर है। (1 marks)
- A और B पर प्रभाव से पहले और बाद में वेग के क्षैतिज और ऊर्ध्वाधर घटक की गणना करें। (2.5 marks)
- उस ऊंचाई की गणना करें जिस पर गेंद दीवार से टकराती है। (0.5 Marks)
- पुनर्प्राप्ति समय (अभ्यास के दौरान खिलाड़ी द्वारा लगातार दो हिट के बीच का समय) (2 marks)

- प्रतिक्रिया R (गेंद और रैकेट के मध्य) की क्षैतिज और ऊर्ध्वाधर घटक, प्रतिक्रिया बल R का परिमाण और प्रतिक्रिया R द्वारा क्षैतिज के साथ बनाया गया कोण। (2 marks)



3. The aircraft landing gear viewed from the front is being retracted immediately after takeoff, and the wheel is spinning at the rate corresponding to the takeoff speed of 200 km/h. The 45-kg wheel has a radius of gyration about its z-axis of 370 mm. Neglect the thickness of the wheel and calculate the angular momentum of the wheel about G where θ is increasing at rate of 30° per second. (5 Marks)

सामने से देखने पर विमान का लैंडिंग गियर टेकऑफ़ के तुरंत बाद वापस लिया जा रहा है, और पहिया 200 किमी/घंटा की टेकऑफ़ गति के अनुरूप दर पर घूम रहा है। 45 किलोग्राम के पहिये की z-अक्ष के चारों ओर घूमने की परिभ्रमण त्रिज्या 370 मिमी है। पहिये की मोटाई की उपेक्षा करें और G के संबंध में पहिये के कोणीय संवेग की गणना करें जहाँ θ , 30° प्रति सेकंड की दर से बढ़ रहा है। (5 Marks)

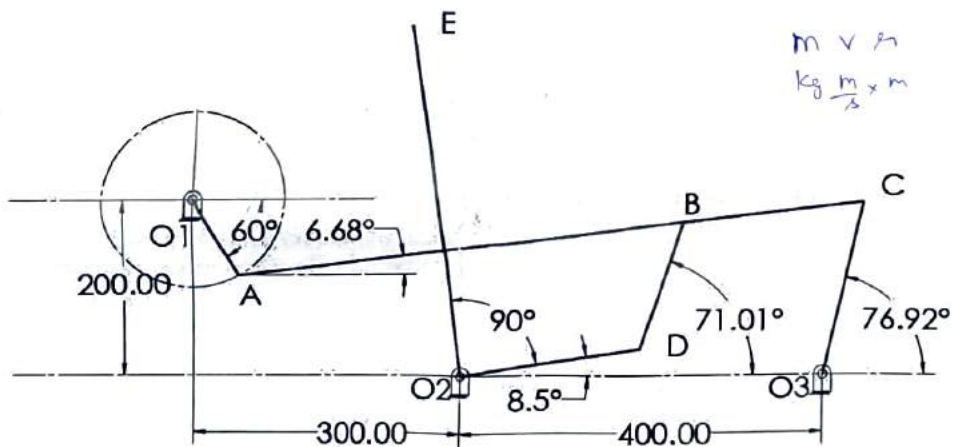
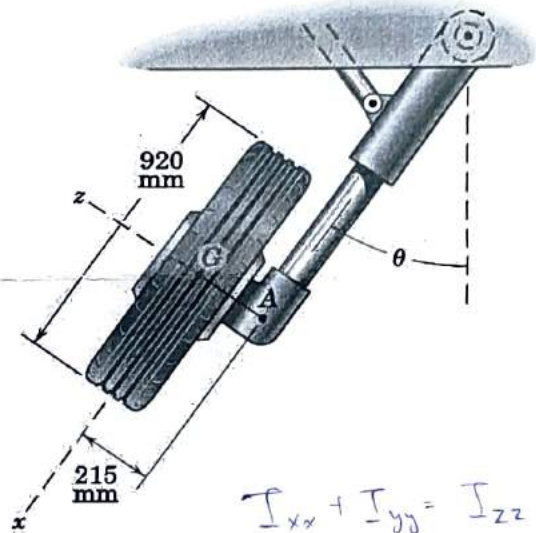


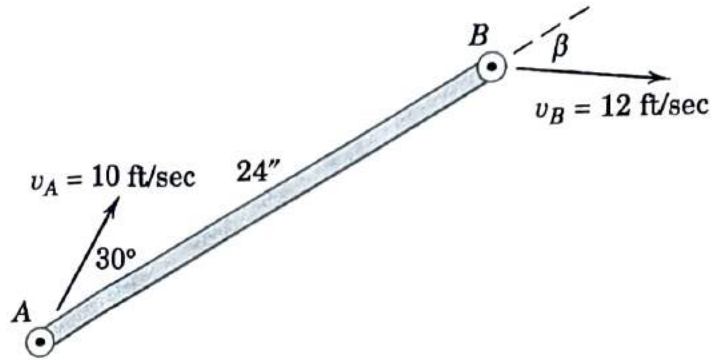
Figure for Question 4

4. The mechanism of a wrapping machine, as shown in the figure, has the following dimensions: $O_1A = 100 \text{ mm}$; $AC = 700 \text{ mm}$; $BC = 200 \text{ mm}$; $O_3C = 200 \text{ mm}$; $O_2E = 400 \text{ mm}$; $O_2D = 200 \text{ mm}$ and $BD = 150 \text{ mm}$. The crank O_1A rotates at a uniform speed of 100 rad/s (clockwise direction). Find the velocity of the point B and point E of the bell crank lever by Analytical Method. (do not use velocity diagram or instantaneous centre method) (10 Marks)

जैसा कि चित्र में दिखाया गया है, रैपिंग मशीन तंत्र के आयाम निम्नलिखित हैं: $O_1A = 100 \text{ mm}$; $AC = 700 \text{ mm}$; $BC = 200 \text{ mm}$; $O_3C = 200 \text{ mm}$; $O_2E = 400 \text{ mm}$; $O_2D = 200 \text{ mm}$ and $BD = 150 \text{ mm}$ । क्रैंक $O_1A = 100 \text{ rad/s}$ (घड़ी की दिशा में) की एक समान गति से घूमता है। विश्लेषणात्मक विधि द्वारा बेल क्रैंक लीवर के बिंदु B और बिंदु E का वेग ज्ञात करें। (वेग आरेख या तात्कालिक केंद्र विधि का उपयोग न करें) (10 Marks)

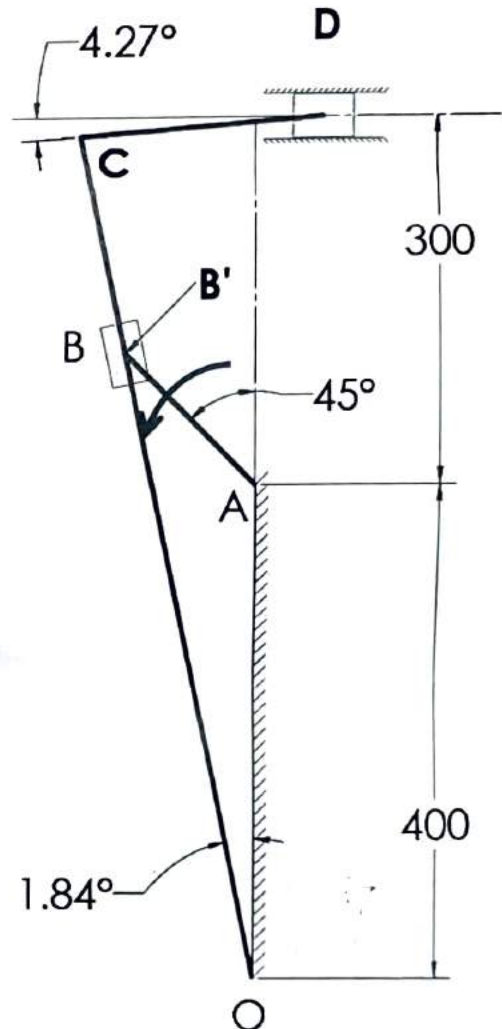
5. End A of the 24-in. link has a velocity of 10 ft/sec in the direction shown. At the same instant end B has a velocity whose magnitude is 12 ft/sec as indicated. Find the relative velocity of B with respect to A and angular velocity of the link. (5 Marks)

24-इंच लिंक के अंत A का वेग दर्शाई गई दिशा में 10 फीट/सेकंड है। उसी समय पर जैसा कि दर्शाया गया है, छोर B के वेग का परिमाण 12 फीट/सेकंड है। A के संबंध में B का सापेक्ष वेग और लिंक का कोणीय वेग ज्ञात करें। (5 Marks)

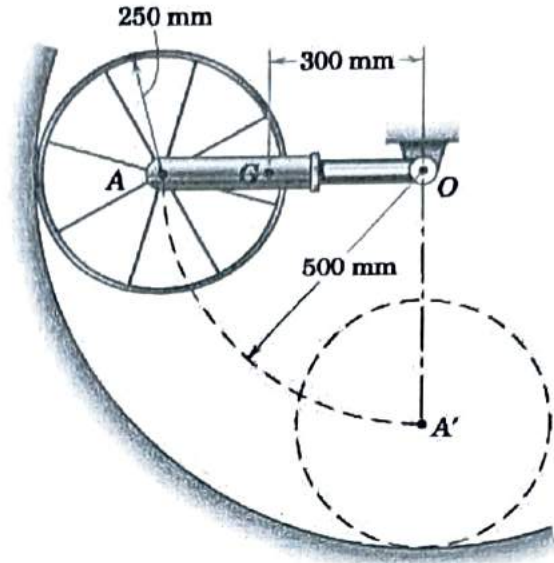


6. A mechanism of a crank and slotted lever quick return motion is shown in the figure. If the crank rotates counterclockwise at 120 r.p.m. , determine for the configuration shown, the velocity and acceleration of the ram D. Also determine the angular acceleration of the slotted lever. Crank, $AB = 150 \text{ mm}$; Slotted arm, $OC = 700 \text{ mm}$ and link $CD = 200 \text{ mm}$. Use Analytical Method only. (do not use velocity diagram or instantaneous centre method) (15 Marks)

चित्र में एक क्रैंक और स्लॉटेड लीवर त्वरित वापसी गति का तंत्र दिखाया गया है। यदि क्रैंक 120 r.p.m. पर वामावर्त घूमता है, तो दिखाए गए क्षण के लिए, रैम D का वेग और त्वरण निर्धारित करें। स्लॉटेड लीवर का कोणीय त्वरण भी निर्धारित करें। क्रैंक, $AB = 150 \text{ मिमी}$; स्लॉटेड आर्म, $OC = 700 \text{ मिमी}$, और लिंक $CD = 200 \text{ मिमी}$ । विश्लेषणात्मक पद्धति का ही प्रयोग करें। (वेग आरेख या तात्कालिक केंद्र विधि का उपयोग न करें) (15 Marks)



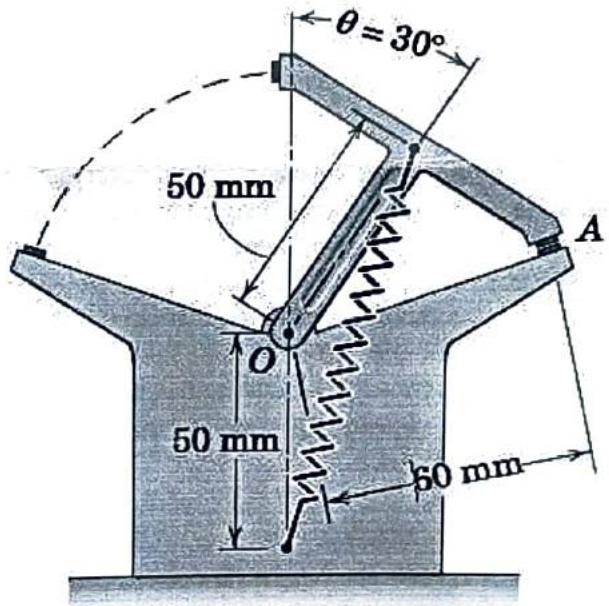
7. The wheel consists of a 4-kg rim of 250 mm radius with hub and its spokes of negligible mass. The wheel is mounted on the 3-kg yoke OA with mass centre at G and with radius of gyration of 350 mm about O. If the assembly is released from rest in the horizontal position shown in the figure and if the wheel rolls on the circular surface without slipping, compute the velocity of point A when it reaches A'. (5 Marks)



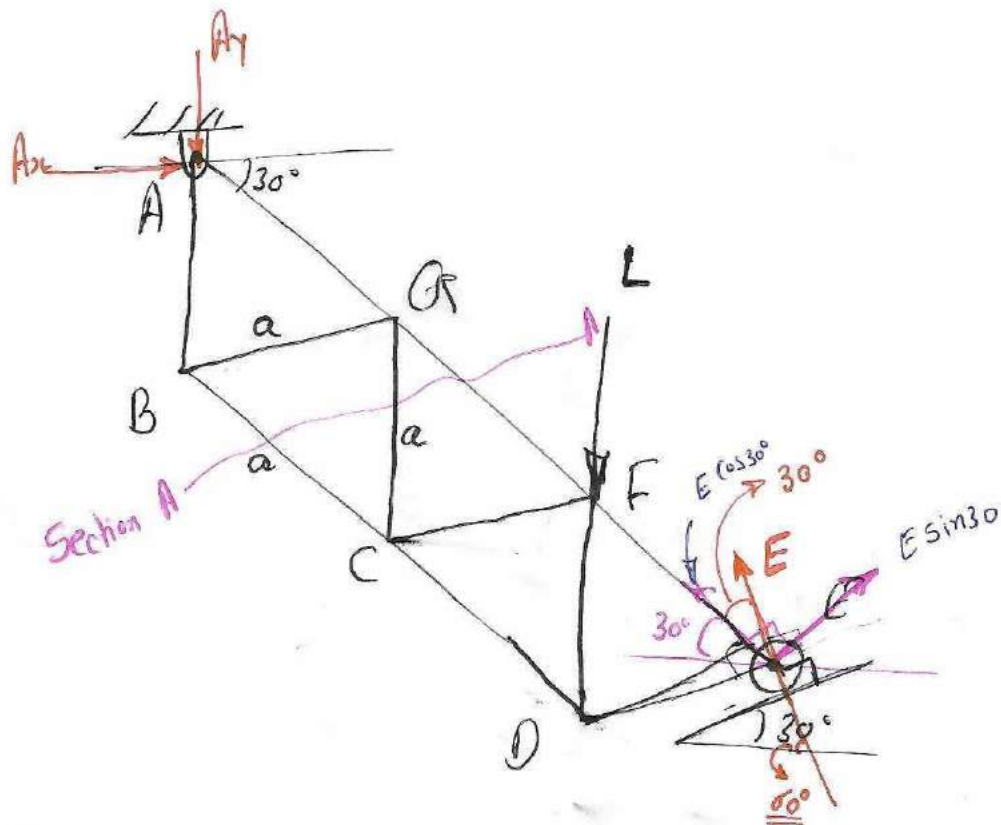
दिया गया पहिया 250 मिमी त्रिज्या का 4 किलोग्राम के रिम, और हब जिसकी तीलियाँ नगण्य द्रव्यमान की हैं, से बना है। पहिया 3 किलोग्राम के योक OA पर लगाया गया है, जिसका द्रव्यमान केंद्र G पर है के और O के संबंध में घुमाव की त्रिज्या 350 मिमी है। यदि असेंबली को चित्र में दिखाई गयी क्षैतिज स्थिति में आराम से छोड़ा जाता है और यदि पहिया गोलाकार सतह पर बिना फिसले घूमता है, बिंदु A के वेग की गणना करें जब यह A' तक पहुँचता है। (5 Marks)

8. Specify the unstretched length l_0 of the spring of stiffness $k = 1400 \text{ N/m}$ which will result in a velocity of 0.25 m/s for the contact at A if the toggle is given a slight nudge from its null position at $\theta = 0$. The toggle has a mass of 1.5 kg and a radius of gyration about O of 55 mm. Motion occurs in the horizontal plane. (5 Marks)

$k = 1400 \text{ N/m}$ कठोरता वाले स्प्रिंग की बिना खींची गई लंबाई l_0 निर्दिष्ट करें, जिसके परिणामस्वरूप A पर संपर्क के लिए 0.25 m/s का वेग होगा यदि टॉगल को $\theta=0$ पर इसकी शून्य स्थिति से थोड़ा सा धक्का दिया जाता है। टॉगल का द्रव्यमान 1.5 किलोग्राम है और O के चारों ओर परिभ्रमण त्रिज्या 55 मिमी है। गति क्षैतिज तल में होती है। (5 Marks)



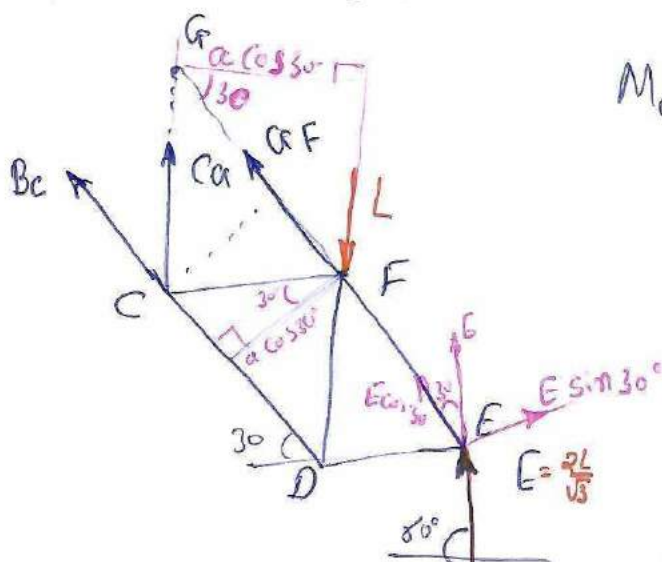
Q-1



$$\sum M_A = 0$$

$$-L \cdot 2a \cos 30^\circ + E \sin 30^\circ (3a) = 0$$

$$\Rightarrow E = \frac{2L}{\sqrt{3}}$$



$$M_A = 0 \Rightarrow -BC \cdot a' \cos 30^\circ + \frac{2L}{\sqrt{3}} \sin 30^\circ \cdot 2a - L \cdot a' \cos 30^\circ = 0$$

$$\Rightarrow BC = \frac{L}{3} \text{ (T)}$$

$$M_F = 0 \Rightarrow -BC \cdot a \cos 30^\circ - CF \cdot a \cos 30^\circ + \frac{2L}{\sqrt{3}} \sin 30^\circ \cdot a = 0$$

$$\Rightarrow BC + CF = \frac{2L}{\sqrt{3}} \tan 30^\circ$$

$$\frac{L}{3} + CF = \frac{2L}{3}$$

$$CF = \frac{L}{3} \text{ (T)}$$

Ans

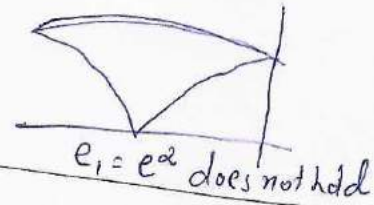
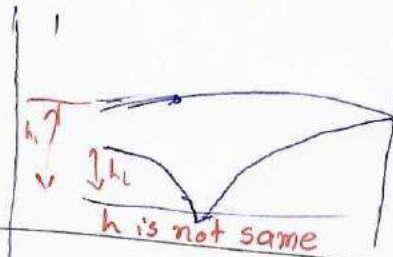
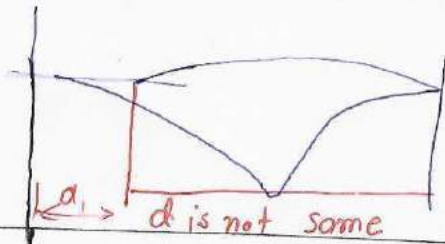
$$E = \frac{2L}{\sqrt{3}}$$

$$BC = \frac{L}{3} \text{ (T)}$$

$$CF = \frac{L}{3} \text{ (T)}$$

3 Marks
2 Value
1 Nature

Q-2 is overconstrained problem So you can omit any one of the following Constraint / if any one of following Constraint is not satisfied still student is entitled to get full Marks according to their findings



$v_{1xc} = v_{0xc}$	24.76 m/s	24.76 m/s	24.76 m/s
$v_{1y} = v_{2y}$	-0.48324 m/s	-0.48324 m/s	-0.48324 m/s
t_{01}	0.404 sec	0.404 sec	0.404 sec
y_1	1.605 m	1.605 m	1.605 m
$v_{2xc} = v_{3xc} = v_{4xc} = v_{5xc}$	-15.35 m/s	-11.93 m/s	-11.93 m/s
v_{3y}	-5.63 m/s	-5.63 m/s	-5.63 m/s
t_{23}	0.525 sec	0.525 sec	0.525 sec
v_{4y}	4.43 m/s	3.71 m/s	4.43 m/s
e_2	0.787	0.66	0.787
e_1	0.62	0.43	0.43
t_{q5}	0.452 sec	0.452 sec	0.452
R_x	50.130 N	45.86 N	45.86 N
R_y	4.60 N	4.60 N	4.60 N
R	50.35 N	46.09 <u>N</u>	46.09 N
β	5.24°	5.72°	5.72°
t	1.379 sec	1.379 <u>sec</u>	1.379 <u>sec</u>

$$V_0 = 25 \text{ m/s}$$

$$d = 10 \text{ m}$$

$$h = 1 \text{ m}$$

$$\theta = 8^\circ$$

$$m = 0.025 \text{ kg or } 25 \text{ gm}$$

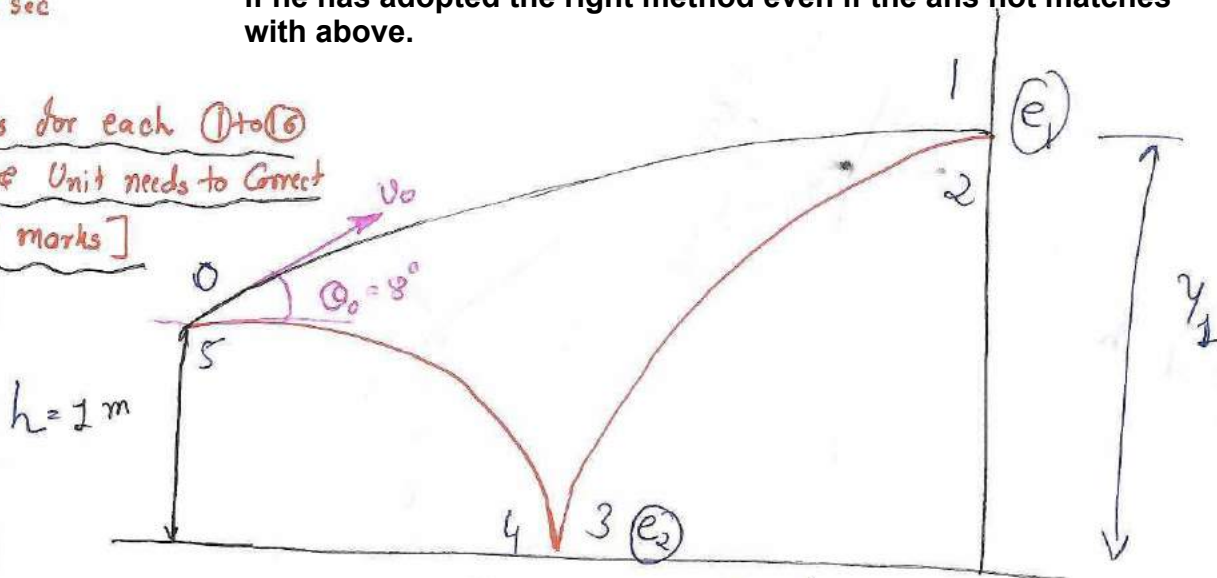
$$\Delta t = 0.02 \text{ sec}$$

Ratio of Coefficient of Restitution of Wall
to Coefficient of Restitution of floor is
Coefficient of Restitution of floor itself.

$$\frac{e_1}{e_2} = e_2 \quad \underline{\underline{Q-2}}$$

This is just a sample solution and student is entitled to get marks if he has adopted the right method even if the ans not matches with above.

* 0.5 Marks for each (1) to (6)
(Both Value & Unit needs to correct
else '0' marks)



[0]

$$V_{0x} = V_0 \cos \theta \approx 24.76 \text{ m/s}$$

$$V_{0y} = V_0 \sin \theta \approx 3.48 \text{ m/s}$$

[1]

$$V_{1x} = V_{0x} = 24.76 \text{ m/s} \quad (1)$$

$$V_{1y} = V_{0y} - g t_{01}$$

$$= 3.48 - 9.81 \times 0.404$$

$$V_{1y} = -0.48324 \text{ m/s} \quad (2)$$

$$t_{01} = \frac{d}{V_{1x}} = \frac{10}{24.76}$$

$$t_{01} \approx 0.404 \text{ sec} \quad (3)$$

$$y_1 = 1 + V_{0y} t_{01} - \frac{1}{2} g t_{01}^2 = 1.605 \text{ m} \quad (4)$$

[2]

$$V_{2y} = V_{1y} = -0.48324 \text{ m/s}$$

$$V_{2x} = -e_1 V_{1x}$$

$$V_{2x} = -15.35 \text{ m/s} \quad (5)$$

[3]

$$V_{3x} = V_{2x}$$

$$V_{3y} = -\sqrt{V_{2y}^2 + 2gy_1}$$

$$V_{3y} = -5.63 \text{ m/s} \quad (6)$$

$$t_{23} = \frac{V_{2y} - V_{3y}}{g}$$

$$t_{23} = 0.525 \text{ sec} \quad (7)$$

[4]

$$V_{4y} = V_{3x}$$

$$V_{4y} = \sqrt{2gh}$$

$$V_{4y} = 4.43 \text{ m/s} \quad (8)$$

Now from

$$V_{4y} = -e_2 V_{3y}$$

$$e_2 = 0.787 \quad (10)$$

from $\frac{e_1}{e_2} = e_2$

$$(11) \quad e_1 \approx 0.62 \quad t_{45} = \frac{V_{4y}}{g}$$

$$t_{45} = 0.452 \text{ sec} \quad (9)$$

[5]

$$V_{5x} = V_{4x}$$

$$V_{5y} = 0 \quad \leftarrow \text{From statement}$$

$$\text{or } V_{5x} \leftarrow V_5 + \frac{m g}{R_x} \int_0^{\Delta t} dt = V_0 \cos 8^\circ$$

Impulse momentum diagram

$$-mV_5 + R_x \Delta t = mV_0 \cos 8^\circ$$

$$R_x = \frac{m}{\Delta t} [V_{5x} + V_5]$$

$$R_x = 50.138 \text{ N} \quad (12)$$

$$R_y = \frac{m}{\Delta t} V_{0y} + mg$$

$$R_y = 4.60 \text{ N} \quad (13)$$

$$R = \sqrt{R_x^2 + R_y^2} = 50.35 \text{ N} \quad (14)$$

$$\beta = \tan^{-1} \left(\frac{R_y}{R_x} \right) = 5.24^\circ \quad (15)$$

Recovery time / time between two consecutive stroke $t = t_{01} + t_{23} + t_{45} = 0.404 + 0.523 + 0.452$
 $t = 1.379 \text{ sec} \quad (16)$

$v = 200 \text{ km/h}$
 $m = 45$
 $k = 370 \text{ mm} \approx 0.37 \text{ m}$

$H_a = ?$ ω is increasing at rate of 30°

$H_n = ?$

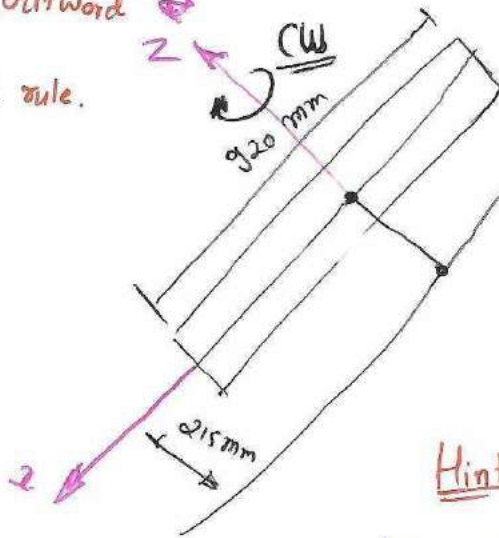
$\Rightarrow \omega_y = -\frac{30 \times \pi}{180} = -0.524 \text{ rad/s}$

Q-3

Y axis is $\perp$ to page / figure plane

Coming outward

As per R.H. rule.



$v = r \omega_2$

$\omega_2 = \frac{v}{r}$

$= \frac{200 \times 10^3}{3600} \times \frac{0.920}{2}$

-ve clockwise rotation of wheel w.r.t. Z axis

$\omega_2 \approx -120.8 \text{ rad/s}$

$\omega_{zc} = 0 \text{ rad/s}$

Hint Since x, y, z is principal axis

$H_a = I_{xx} \omega_x \hat{i} + I_{yy} \omega_y \hat{j} + I_{zz} \omega_z \hat{k}$

$I_{zz} = mk^2 = 45 \times 0.37^2 = 6.16 \text{ kg m}^2$ (1) Marks $\rightarrow$ 0.5 Value
 0.5 Unit

From perpendicular Axis theorem - $I_{zz} = I_{xx} + I_{yy}$

For disc with negligible thickness $\Rightarrow I_{xx} = I_{yy}$

$I_{xx} = I_{yy} = 3.08 \text{ kg m}^2$ $\rightarrow$ (1) mark $\rightarrow$ 0.5 Value
 0.5 Unit

$H_a = 3.08 (0) \hat{i} + 3.08 (-0.524) \hat{j} + 6.16 (-120.8) \hat{k}$

$H_a = -1.613 \hat{j} - 744.13 \hat{k} \text{ kg m}^2/\text{s}$

(2) Marks $\rightarrow$ 1 for value
 $\rightarrow$ 0.5 for direction
 $\rightarrow$ 0.5 for Unit

(1) marks $\rightarrow$ ω_y $\rightarrow$ 0.5 marks $\rightarrow$ For value + Unit (Correct)
 $\rightarrow$ 0 marks $\rightarrow$ If value and/or Unit is wrong
 $\rightarrow$ ω_z $\rightarrow$ 0.5 marks $\rightarrow$ For value + Unit (Correct)
 $\rightarrow$ 0 marks $\rightarrow$ If value and/or Unit is wrong.

$$\omega = 100 \text{ rad/s} \quad \text{CW} = -100 \hat{k}$$

$$V_E = (?)$$

Q4

$$\vec{O_1A} = 100 \cos 60^\circ \hat{i} - 100 \sin 60^\circ \hat{j}$$

$$\vec{AC} = 700 \cos 6.68^\circ \hat{i} + 700 \sin 6.68^\circ \hat{j} = -\vec{CA}$$

$$\vec{AB} = 500 \cos 6.68^\circ \hat{i} + 500 \sin 6.68^\circ \hat{j} = -\vec{BA}$$

$$\vec{BC} = 200 \cos 6.68^\circ \hat{i} + 200 \sin 6.68^\circ \hat{j} = -\vec{CB}$$

$$\vec{BD} = -150 \cos 71.01^\circ \hat{i} + -150 \sin 71.01^\circ \hat{j} = -\vec{DB}$$

$$\vec{O_2D} = 200 \cos 8.5^\circ \hat{i} + 200 \sin 8.5^\circ \hat{j} = -\vec{DO_2}$$

$$\vec{O_2E} = 400 \cos 98.5^\circ \hat{i} + 400 \sin 98.5^\circ \hat{j}$$

$$\vec{O_3E} = 200 \cos 76.92^\circ \hat{i} + 200 \sin 76.92^\circ \hat{j}$$

$$\vec{V}_A = \vec{V}_0 + \omega_A \times \vec{O_1A} + \vec{V}_{rel}$$

$$\vec{V}_A = -100 \hat{k} \times (50\hat{i} - 86.60\hat{j})$$

$$\vec{V}_A = -8660\hat{i} - 5000\hat{j} \text{ mm/s}$$

$$\vec{V}_E = \vec{V}_A + \omega_{AC} \times \vec{AC} = \vec{V}_{O_3} + \omega_{O_3C} \times \vec{O_3C}$$

$$= -8660\hat{i} - 5000\hat{j} + \omega_{AC} \hat{k} \times (695.25\hat{i} + 81.43\hat{j})$$

$$= \omega_{O_3C} \hat{k} \times [45.26\hat{i} + 194.81\hat{j}]$$

$$= [-8660 - 84.43\omega_{AC}]\hat{i} + [-5000 + 695.25\omega_{AC}]\hat{j}$$

$$= -194.81\omega_{O_3C}\hat{i} + 45.26\omega_{O_3C}\hat{j}$$

$$84.43\omega_{AC} - 194.81\omega_{O_3C} = -8660 \quad (\hat{i} \text{ equation})$$

$$695.25\omega_{AC} - 45.26\omega_{O_3C} = 5000 \quad (\hat{j} \text{ equation})$$

or $\omega_{AC} = 10.378 \text{ k rad/s}$

$$\omega_{AC} \approx 10.378 \text{ k rad/s}$$

(CCW)

$$\omega_{O_3C} \approx 48.952 \text{ k rad/s}$$

(CCW)

$$\vec{V}_B = \vec{V}_A + \omega_{AC} \times \vec{AB}$$

$$= -8660\hat{i} - 5000\hat{j} + 10.378 \hat{k} \times [496.61\hat{i} + 58.16\hat{j}]$$

$$\vec{V}_B = -9263.584\hat{i} + 153.819\hat{j} \text{ mm/s}$$

$$V_B = 9264.86 \text{ mm/s}$$

$$\theta = 179^\circ$$

3 Marks

1.5 value

0.5 Unit

1 direction

$$\vec{O_1A} = 50\hat{i} - 86.60\hat{j} / 1000$$

$$\vec{AC} = 695.25\hat{i} + 81.43\hat{j} / 1000$$

$$\vec{AB} = 496.61\hat{i} + 58.16\hat{j} / 1000$$

$$\vec{BC} = 198.64\hat{i} + 23.27\hat{j} / 1000$$

$$\vec{BD} = -48.81\hat{i} - 141.84\hat{j} / 1000$$

$$\vec{O_2D} = 197.80\hat{i} + 29.56\hat{j} / 1000$$

$$\vec{O_2E} = -59.12\hat{i} + 395.61\hat{j} / 1000$$

$$\vec{O_3E} = 45.26\hat{i} + 194.81\hat{j} / 1000$$

$$\vec{V}_D = \vec{V}_B + \omega_{BD} \times \vec{BD} = \vec{V}_{O_2} + \omega_{O_2D} \times \vec{O_2D}$$

$$-9263.584\hat{i} + 153.819\hat{j} + \omega_{BD} \hat{k} \times [-48.81\hat{i} - 141.84\hat{j}]$$

$$= \omega_{O_2D} \hat{k} \times [197.80\hat{i} + 29.56\hat{j}]$$

$$-9263.584\hat{i} + 141.84\omega_{BD}\hat{i} + 153.819\hat{j} - 48.81\omega_{BD}\hat{j}$$

$$= -29.56\omega_{O_2D}\hat{i} + 197.80\omega_{O_2D}\hat{j}$$

$$141.84\omega_{BD} + 29.56\omega_{O_2D} = 9263.584 \quad (\hat{i} \text{ equation})$$

$$197.80\omega_{O_2D} + 48.81\omega_{BD} = 153.819 \quad (\hat{j} \text{ equation})$$

$$\omega_{BD} \approx 68.68 \text{ k rad/s} \quad (\text{CCW})$$

$$\omega_{O_2D} \approx -16.17 \text{ k rad/s} \rightarrow [16.17 \text{ rad/s CW}]$$

2 Marks - 1. Value, 0.5 Unit, 0.5 direction

$$\vec{V}_E = \vec{V}_{O_2} + \omega_{O_2D} \times \vec{O_2E} + \vec{V}_{rel}$$

$$= -16.17 \hat{k} \times [-59.12\hat{i} + 395.61\hat{j}]$$

$$\vec{V}_E = 6397.014\hat{i} + 955.970\hat{j} \text{ mm/s}$$

$$\vec{V}_E = 6468.05 \text{ mm/s}$$

$$\vec{V}_E = 6.468 \text{ m/s}$$

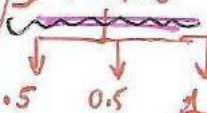
$$\theta = 8.5^\circ \text{ with Horizontal}$$

1.5 Value

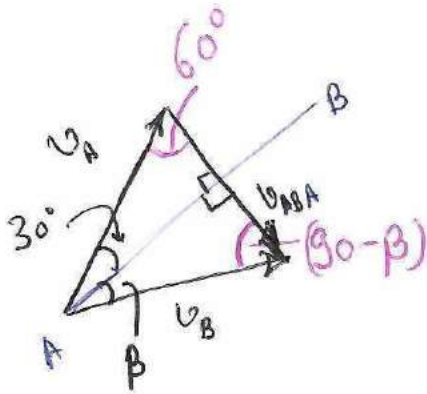
0.5 Unit

1 direction

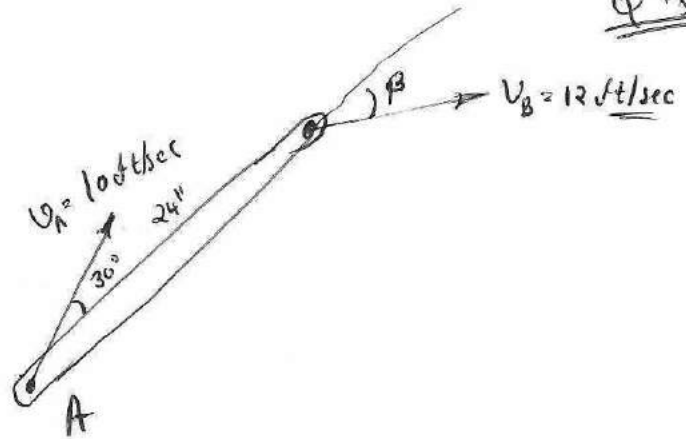
3 Marks



Q5



Q.5



Hint $v_{AB} \perp AB$ (link)

From Sine Law

$$\frac{v_A}{\sin(90-\beta)} = \frac{v_B}{\sin 60^\circ}$$

$$\cos \beta = \frac{v_A}{v_B} \sin 60^\circ$$

$$\boxed{\beta = 43.81^\circ}$$

$$\frac{v_B}{\sin 60} = \frac{v_{BA}}{\sin 73.81}$$

$$\boxed{v_{BA} = 13.306 \text{ m/s}}$$

2.5 Marks $\left\{ \begin{array}{l} \rightarrow 1.5 \text{ Value} \\ \rightarrow 1 \text{ Unit} \end{array} \right.$

$$v_{BA} = r \omega_{BA}$$

$$\omega_{BA} = \frac{13.306}{2}$$

$$(\because 2 \text{ ft} = 12 \text{ in} \Rightarrow 24 \text{ in} = 2 \text{ ft})$$

$$\boxed{\omega_{BA} = 6.653 \text{ rad/sec (CW)}}$$

2.5 Marks $\left\{ \begin{array}{l} \rightarrow 1.5 \text{ Value} \\ \rightarrow 0.5 \text{ Unit} \\ \rightarrow 0.5 \text{ direction} \end{array} \right.$

$$AB = 150 \text{ mm}$$

$$OC = 700 \text{ mm}$$

$$CD = 200 \text{ mm}$$

$$\omega = 120 \text{ rpm ccw} \rightarrow 12.57 \text{ rad/s ccw}$$

$$\text{from } \triangle OAB \quad \angle R = 135^\circ$$

$$\cos 135^\circ = \frac{AB^2 + OA^2 - OB^2}{2AB \times OA}$$

$$OB = 517.06 \text{ mm}$$

$$\vec{AB} = -150 \sin 45^\circ \hat{i} + 150 \cos 45^\circ \hat{j}$$

$$\vec{OB} = -517.06 \sin 11.84^\circ \hat{i} + 517.06 \cos 11.84^\circ \hat{j}$$

$$\vec{OC} = -700 \sin 11.84^\circ \hat{i} + 700 \cos 11.84^\circ \hat{j}$$

$$\vec{CD} = 200 \cos 4.22^\circ \hat{i} + 200 \sin 4.22^\circ \hat{j}$$

$$\vec{V}_B = \vec{V}_A + \omega_{BA} \times \vec{AB} + \vec{V}_{rel} = \vec{V}_O + \omega_{BO} \times \vec{OB} + \vec{V}_{rel}$$

$$\therefore 12.57 \text{ k} \times [-106.07 \hat{i} + 106.07 \hat{j}] = \omega_{BO} \times [-106.09 \hat{i} + 506.06 \hat{j}] + \vec{V}_{BB'}$$

$$-1333.30 \hat{i} - 1333.30 \hat{j} = [-506.06 \omega_{BO} + 0.205 \vec{V}_{BB'}] \hat{i} + [-106.09 \omega_{BO} + 0.979 \vec{V}_{BB'}] \hat{j}$$

$$-506.06 \omega_{BO} + 0.205 \vec{V}_{BB'} = -1333.30$$

$$-106.09 \omega_{BO} + 0.979 \vec{V}_{BB'} = -1333.30$$

$$\omega_{BO} = 2.1786 \text{ k rad/s [ccw]}$$

$$\vec{V}_{BB'} = -1125.8 \text{ mm/s}$$

$$\Rightarrow \vec{V}_{BB'} = -$$

$$\vec{V}_C = \vec{V}_O + \omega_{CO} \times \vec{OC} + \vec{V}_{rel}$$

$$= 3.05 \times [-143.63 \hat{i} + 685.11 \hat{j}]$$

$$\vec{V}_C = -2089.586 \hat{i} - 438.071 \hat{j} \text{ mm/s}$$

$$|\vec{V}_C| = V_C = 2135.01 \text{ mm/s}$$

$$\approx 2.135 \text{ m/s}$$

$$\vec{V}_D = (?)$$

$$\vec{a}_D = (?)$$

$$\alpha_{OC} = (?) = \text{Value} + \text{Unit} + \text{direction} - 1 + 0.5 + 0.5$$

$$\omega_{BO} \rightarrow \text{Value} + \text{Unit} + \text{direction} - 1 - 1$$

$$|\vec{V}_{rel}| = \vec{V}_{BB'} \rightarrow \text{Value} + \text{Unit} + \text{direction} - 1 - 1$$

$$\vec{V}_D \rightarrow \text{Value} + \text{Unit} + \text{direction} - 1 + 0.5 + 0.5$$

$$\alpha_{BO} \rightarrow \text{Value} + \text{Unit} + \text{direction} - 1 - 1$$

$$\vec{a}_{BB'} = \vec{a}_{rel} \rightarrow \text{Value} + \text{Unit} + \text{direction} - 1 + 0.5 + 0.5$$

$$a_D \rightarrow \text{Value} + \text{Unit} + \text{Direction} - 2 + 0.5 + 0.5$$

$$\vec{a}_C \rightarrow \text{Value} + \text{Unit} + \text{Direction} - 1 + 0.5 + 0.5$$

$$\vec{AB} = -106.07 \hat{i} + 106.07 \hat{j}$$

$$\vec{OB} = -106.09 \hat{i} + 506.06 \hat{j}$$

$$\vec{OC} = -143.63 \hat{i} + 685.11 \hat{j}$$

$$\vec{CD} = 199.45 \hat{i} + 14.89 \hat{j}$$

Direction of $\vec{V}_{BB'}$ is || to $\vec{OB}$

$$\Rightarrow \vec{V}_{BB'} = V_{BB'} [\sin 11.84^\circ \hat{i} + \cos 11.84^\circ \hat{j}]$$

$$\vec{V}_{BB'} = V_{BB'} [\sin 11.84^\circ \hat{i} + \cos 11.84^\circ \hat{j}]$$

Value + Unit + direction

$$\omega_{BO} = 3.05 \text{ k rad/s [ccw]}$$

$$\vec{V}_{BB'} = -1031.1 \text{ mm/s} \approx 1.03 \text{ m/s}$$

$$\vec{V}_{BB'} = 211.39 \hat{i} - 1009.45 \hat{j} \text{ mm/s}$$

2 Marks
Value + Unit + direction

$$\vec{V}_D = \vec{V}_C + \omega_{CO} \times \vec{CD} + \vec{V}_{rel}$$

$$= -2089.586 \hat{i} - 438.071 \hat{j} + \omega_{CO} \times [199.45 \hat{i} + 14.89 \hat{j}]$$

$$= [-2089.586 - 14.89 \omega_{CO}] \hat{i} + [-438.071 + 199.45 \omega_{CO}] \hat{j}$$

Since D can slide Horizontally Vertical Component of $\vec{V}_D$ must be zero $\Rightarrow \vec{V}_D \cdot \hat{j} = 0$

$$-438.071 + 199.45 \omega_{CO} = 0 \Rightarrow \omega_{CO} = 2.20 \text{ k rad/s [ccw]}$$

$$\vec{V}_D = -2122.34 \hat{i} \text{ mm/s}$$

$$\vec{V}_D = -2122.34 \hat{i} \text{ mm/s} \approx -2.12 \hat{i} \text{ m/s}$$

2 Marks
1 - Value
0.5 Unit
0.5 direction

$$\vec{a}_B = \vec{a}_O + \alpha_{OB} \times \vec{OB} + \omega_{OB} \times (\omega_{OB} \times \vec{OB}) + 2\omega_{OB} \times \vec{v}_{rel} + \vec{a}_{rel}$$

$$\vec{a}_B = \vec{a}_A + \alpha_{BA} \times \vec{AB} + \omega_{BA} \times (\omega_{BA} \times \vec{AB}) + 2\omega_{BA} \times \vec{v}_{rel} + \vec{a}_{rel} = \vec{a}_O + \alpha_{OB} \times \vec{OB} + \omega_{OB} \times (\omega_{OB} \times \vec{OB}) + 2\omega_{OB} \times \vec{v}_{rel} + \vec{a}_{rel}$$

$$\therefore \frac{12.57}{3.05} \hat{k} \times \left[\frac{12.57}{3.05} \hat{k} \times (-106.07 \hat{i} + 106.07 \hat{j}) \right] = \alpha_{OB} \hat{k} \times [-106.09 \hat{i} + 506.06 \hat{j}] + 3.05 \hat{k} \times [3.05 \hat{k} \times (-106.09 \hat{i} + 506.06 \hat{j})]$$

$$+ 2 \times 3.05 \hat{k} \times [211.38 \hat{i} - 1009.45 \hat{j}] + \vec{a}_{rel}$$

① Mark (Value+Unit+direction)

$$\vec{a}_{rel} = a_{rel} [-0.205 \hat{i} + 0.979 \hat{j}]$$

$$\begin{aligned} 506.06 \alpha_{OB} + 0.205 a_{rel} &= -9615.032 \\ -106.09 \alpha_{OB} + 0.979 a_{rel} &= -13340.795 \end{aligned} \quad \text{Solving} \rightarrow \quad \alpha_{OB} = -12.9 \text{ k rad/s}^2$$

$$12.9 \text{ rad/s}^2 \text{ (CW)}$$

$$a_{rel} = -15026 \text{ mm/s}^2$$

② Marks - 1 unit+value
0.5 Unit
0.5 direction

$$\vec{a}_{BB'}$$

$$\vec{a}_{rel} = -3080.33 \hat{i} + 14710.454 \hat{j} \text{ mm/s}^2$$

$$\vec{a}_C = \vec{a}_O + \alpha_{CO} \times \vec{OC} + \omega_{CO} \times (\omega_{CO} \times \vec{OC}) + 2\omega_{CO} \times \vec{v}_{rel} + \vec{a}_{rel}$$

$$\alpha_{CO} = \alpha_{BO} \cdot d_{OB}$$

$$= -12.9 \text{ k} \times [-143.63 \hat{i} + 685.11 \hat{j}] + 3.05 \hat{k} \times [3.05 \hat{k} \times (-143.63 \hat{i} + 685.11 \hat{j})]$$

② Marks - 1 value 0.5 unit 0.5 direction

$$\vec{a}_C = 10174.036 \hat{i} - 4520.41 \hat{j} \text{ mm/s}^2 \Rightarrow |\vec{a}_C| = 11133.06 \text{ mm/s}^2 \approx 11.133 \text{ m/s}^2$$

$$\vec{a}_D = \vec{a}_C + \alpha_{DC} \times \vec{CD} + \omega_{DC} \times (\omega_{DC} \times \vec{CD}) + 2\omega_{DC} \times \vec{v}_{rel} + \vec{a}_{rel}$$

$$= 10174.036 \hat{i} - 4520.41 \hat{j} + \alpha_{DC} \hat{k} \times [199.45 \hat{i} + 14.89 \hat{j}] + 2 \cdot 2 \hat{k} \times [2 \hat{k} \times (199.45 \hat{i} + 14.89 \hat{j})]$$

$$= [10174.036 - 14.89 \alpha_{DC} - 965.338] \hat{i} + [-4520.41 + 199.45 \alpha_{DC} - 72.068] \hat{j}$$

Hint → Since slider D can only move in Horizontal direction only the Component of $\vec{a}_D$ in vertical direction will be zero $\Rightarrow \vec{a}_C \cdot \hat{j} = 0$

$$-4520.41 + 199.45 \alpha_{DC} - 72.068 = 0 \Rightarrow \alpha_{DC} = 23.026 \text{ rad/s}^2 \text{ (CCW)}$$

$$\vec{a}_D = 8865.85 \hat{i} \text{ mm/s}^2$$

or

$$8.866 \text{ m/s}^2$$

③ marks → 2 Value
0.5 Unit
0.5 Direction

$$\text{or } \alpha_{DC} = 23.026 \text{ k rad/s}^2$$

② Marks - 2 value
0.5 Unit
0.5 direction

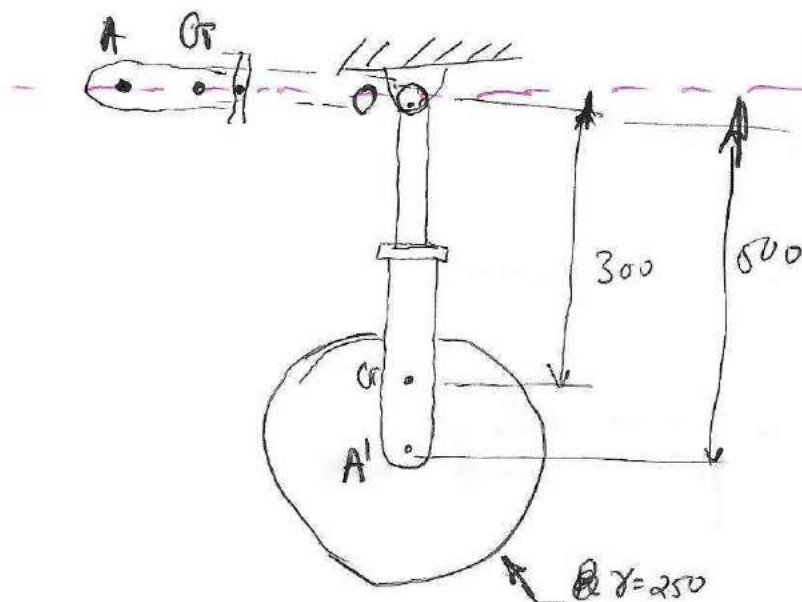
$$m_y = 4 \text{ kg}$$

$$r_y = 250 \text{ mm}$$

$$m_y = 3 \text{ kg}$$

$$r_y = 300 \text{ mm}$$

$$r_y = 0.350 \text{ m}$$



Q-7
Reference

Now as per Energy Conservation

$$K.E._1 + P.E._1 = K.E._2 + P.E._2$$

1- A position
2- A' position

$$\therefore (K.E._2 - K.E._1) + (P.E._2 - P.E._1) = 0$$

$$\therefore \Delta K.E. + \Delta P.E. = 0 \quad \text{--- (1)}$$

Considering Horizontal line passing through O as reference datum.

$$\Delta P.E._2 = \Delta P.E._{yoke} + \Delta P.E._{rim}$$

$$= 3 \times 9.81 \times (-0.3) + 4 \times 9.81 \times (-0.5)$$

$$= -8.83 + -19.62$$

(P.E. = mgh)

$$\Delta P.E. = -28.45 \text{ J}$$

$$\Delta K.E._2 = \Delta K.E._{yoke} + \Delta K.E._{rim}$$

$$= \frac{1}{2} I_o \omega^2 + \frac{1}{2} I_c \omega^2$$

$$= \frac{1}{2} \times (3 \times 0.35^2) \left(\frac{v_A}{0.5} \right)^2 + \frac{1}{2} (8 \times 0.25^2) \left(\frac{v_A}{0.25} \right)^2$$

$$\Delta K.E. = 4.735 v_A^2$$

$$\left. \begin{aligned} I_c &= I_o + m d^2 \\ &= 8 \\ &= 4 \times 0.25^2 + 4 \times 0.25^2 \\ &= 8 \times 0.25^2 \end{aligned} \right\}$$

Putting values in ①

$$4.735 v_A^2 = 28.45$$

$$v_A = 2.45 \text{ m/s}$$

Rubrics

~~KE_{1y}~~ = 0

2 Mark Each — 4 Marks
↓ 0.5 Value, 0.5 Unit

$KE_{1y} = 0 \text{ J}$	$KE_{2y} = 0.735 v_A^2 \text{ J}$
$KE_{1x} = 0 \text{ J}$	$KE_{2x} = 4 v_A^2 \text{ J}$
$PE_{1x} = 0 \text{ J}$	$PE_{2x} = -19.62 \text{ J}$
$PE_{1y} = 0 \text{ J}$	$PE_{2y} = -8.83 \text{ J}$

$$\Delta KE_y = 0.735 v_A^2 \text{ J}$$

$$\Delta KE_x = 4 v_A^2 \text{ J}$$

$$\Delta PE_x = -19.62 \text{ J}$$

$$\Delta PE_y = -8.83 \text{ J}$$

Or

4 Marks ← 2 Marks each
1.5 Value, 0.5 Unit

$$\Delta KE = 4.735 v_A^2 \text{ J}$$

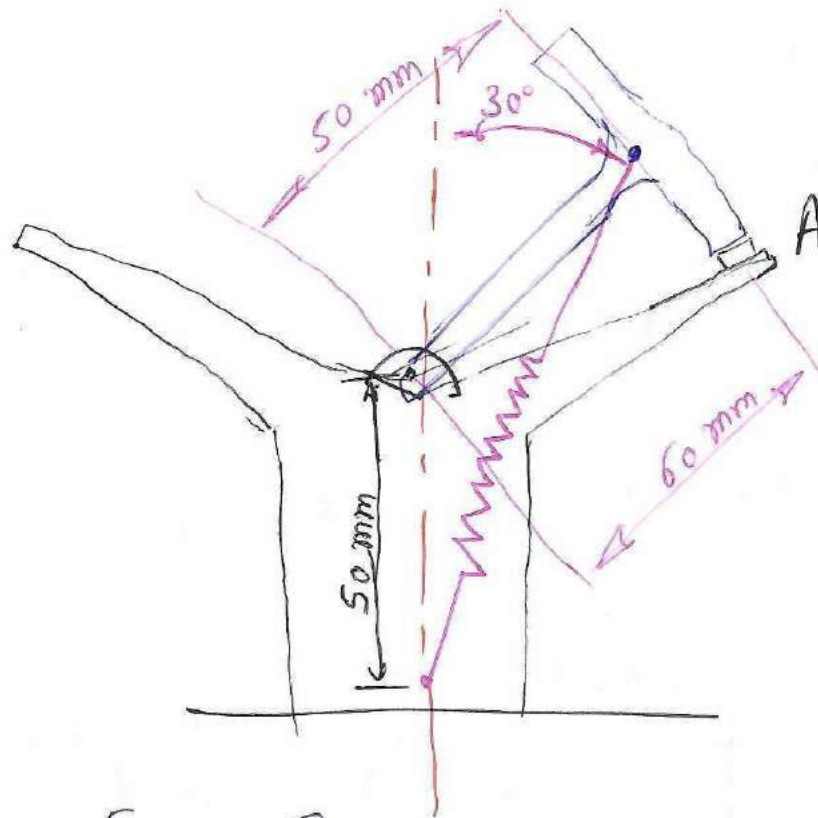
$$\Delta PE = -28.45 \text{ J}$$

2 Mark → 0.5 Value
0.5 Unit

$$v_A = 2.45 \text{ m/s}$$

Q8

$k = 1400 \text{ N/m}$
 $v = 0.25 \text{ m/s}$
 $m = 1.5 \text{ kg}$
 $\bar{k} = 55 \text{ mm}$



Now $\Rightarrow$ Based on Energy Conservation Law

$$PE_1 + KE_1 = PE_2 + KE_2 \quad \text{--- (1)}$$

$1 \rightarrow \theta = 0^\circ$
 $2 \rightarrow \text{Impact at } A$
 $\theta = 30^\circ$

$$P.E._1 = \frac{1}{2} k x^2$$

1 Mark (0.5 value)
(0.5 unit)

$$P.E._1 = \frac{1}{2} k (0.1 - l_0)^2 \text{ J}$$

$$= 700 (0.1 - l_0)^2 \text{ J}$$

Where x change in length
 $= (l_1 - l_0)$

Now at $\theta = 0^\circ$ the mechanism is at rest $\Rightarrow$

$$KE_1 = 0 \text{ J}$$

0.5 Marks

$$l_1 = 0.05 + 0.05$$

$$l_1 = 0.1 \text{ m}$$

~~0.1 m~~

For K.E.2

$$= \frac{1}{2} I_0 \omega^2 + \frac{1}{2} m v^2$$

$$= \frac{1}{2} \times 1.5 \times 0.055^2 \times \frac{0.25^2}{0.06^2} + \frac{1}{2} \times 1.5 \times 0.25^2$$

$I_0 = m \bar{k}^2$ - radius of gyration not stiffness

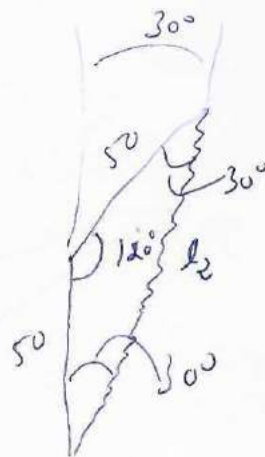
$$K.E._2 = 86.263 \times 10^{-3} \text{ J}$$

1 Marks (0.5 value)
(0.5 unit)

Using

For P.E.₂ = $\frac{1}{2} k x^2$

= $\frac{1}{2} k (l_2 - l_0)^2$



l_2 is calculated using sine law as

$$\frac{0.05}{\sin \frac{30^\circ}{15}} = \frac{l_2}{\sin \frac{120^\circ}{150}}$$

$$l_2 = \frac{86.603 \times 10^{-3} \text{ m}}{96.60}$$

0.5 marks (with correct)
Unit
0 - value/unit wrong.

$$\therefore PE_2 = 700 (86.603 \times 10^{-3} - l_0)^2$$

① Marks
0.5 Unit
0.5 value

Putting Values in equation (1)

$$0 + 700 (0.1 - l_0)^2 = 86.263 \times 10^{-3} + 700 (86.603 \times 10^{-3} - l_0)^2$$

$$\therefore 700 \left[\underbrace{(0.1 - l_0)^2}_{a^2} - \underbrace{(86.603 \times 10^{-3} - l_0)^2}_{b^2} \right] = 86.263 \times 10^{-3}$$

$$[a^2 - b^2 = (a-b)(a+b)]$$

$$\therefore \frac{0.003407}{0.003407} (0.196603 - 2l_0) = \frac{86.263 \times 10^{-3}}{700}$$

$$\Rightarrow l_0 = \frac{80.21}{88.212} \times 10^{-3} \text{ mm}$$

$$\approx 80 \text{ mm}$$

① Marks
0.5 Value
0.5 Unit